M. Tech. and M.Tech-PhD

in

Civil and Infrastructure Engineering with Major in Energy

Introduction

With the tremendous focus on energy sources such as solar, wind, hydropower to name a few, skills to develop necessary infrastructure for energy generation, transmission and distribution from these sources are required. To cater to future needs and emerging infrastructure, the M.Tech. and M. Tech-Ph. D. Program on Civil and Infrastructure Engineering with a major (specialization) in "Energy" is exclusively designed. The program is offered to the students/candidates with diverse engineering backgrounds from India and abroad who are interested in learning and applying the fundamental and advanced technologies (energy conversion, transmission, distribution, storage and conservation) in different areas of energy engineering. The program equips students with a unique set of skills desired in multidisciplinary and dynamic energy industries. To facilitate prospective graduates with the scope of this program, "Career Trajectory" maps for all disciplines are developed. A "Career Trajectory" is a framework for understanding the demand of different energy industries in terms of skills and their mapping with the courses that are being offered in this program across multiple departments. It will assist students in identifying the core skills (general, core, and advanced) and courses to build their career in the chosen energy industry.

The program constitutes a unique value-added model that allows students to select courses from a) "Specialized Stream Electives" and (b) "M. Tech Electives" buckets to build their knowledge in a specialized stream and gain skills they would need to enter into the desired energy industry. Specialized courses are designed to tap rapidly changing energy market while employing emerging technologies. Furthermore, it builds a platform for multidisciplinary research to unravel complex problems associated with global energy infrastructure. After successful completion of the program, students will be able to continue to a Doctor of Philosophy (Ph.D.) in a number of related areas of chosen stream. This program potentially delivers an ability to the graduates to change the landscape of our planet through advanced energy generation and management practices.
An outline of the concept behind this program is presented in the following sections.

Objective of the Program

IIT Jodhpur Civil and Infrastructure Engineering graduates with a major in Energy will:

1. Enter the industry as professionals and leaders to implement solutions to diverse technical challenges of energy infrastructure.
2. Pursue innovative and advanced research as scholars and scientists in leading industry and academic R&D centers around the world to develop state-of-the-art technologies related to future energy infrastructure.
3. Establish and lead startups in emerging areas of sustainability, resilience, management, and effective utilization of various sources of energy.
4. Frame policies, management regulations and ethical guidelines correlating dynamic energy demands, utilization and distribution.

Graduate attributes

A successful graduate should be able to:

1. Understand the relation between different engineering aspects (i.e. electrical, mechanical, chemical, etc.) of various energy infrastructures.
2. Understand the impacts and limitations of different energy resources under various conditions.

3. Understand and relate the engineering, economy, and policy aspects of energy utilization.
4. Design and implement energy-efficient infrastructure related to generation, transmission, distribution of energy.
5. Carry out in-depth multi-faced investigations (e.g. analytical, experimental, and numerical) in various infrastructure engineering issues and propose innovative solutions.
6. Design new models, collect relevant data, and perform validation to help in decision making, developing and implementing advanced energy infrastructure concepts, and solving engineering issues.
7. Participate and work in interdisciplinary teams, demonstrate knowledge of advanced management skills, and implement sustainable practices in infrastructure engineering projects.

Learning outcome

Upon successful completion of the program, the graduate will have an ability

1. To apply and demonstrate acquired knowledge of popular (solar, wind, etc.) and upcoming (tidal, biomass, fuel cell, hydrogen, etc.) energy technologies and their operation in infrastructure engineering practices.
2. To apply engineering design and management aspects of the complete lifecycle of energy: from generation to distribution.
3. To identify, formulate and apply basics of energy harvesting and storage to mitigate challenges related to power fluctuations.
4. To apply appropriate techniques of modeling and prediction, modern engineering and IT tools to complex engineering activities with an understanding of the limitations.
5. To plan, design and execute large scale infrastructure projects on smart cities to meet future energy needs.
6. To compete, lead, and practice a successful professional career in the core areas of expertise with specialized knowledge in the interdisciplinary energy infrastructure sector.

Overall Structure (M. Tech. in Civil and Infrastructure Engineering with Major in Energy)

Category	Program Core (PC)	Specialization Stream Electives (SSE)	M. Tech. Elective (ME)	M. Tech. Open (MO)	M. Tech. Project (MP)	Non-Graded Courses	Total Graded
Credits	**3**	**17**	**17**	**3**	**16**	**4**	**60**

Semester wise distribution of credits

Cat.	*Course Title*	*L-T-P*	*Credits*	*Cat.*	*Course Title*	*L-T-P*	*Credits*
I Semester				**II Semester**			
PC	*Program Core*	3-0-0	3	*SSE*	*Specialization stream - 5*	3-0-0	3
SSE	*Specialization stream -1*	3-0-0	3	*SSE*	*Specialization stream - 6*	0-0-4	2
SSE	*Specialization stream -2*	3-0-0	3	ME	*Program Elective – 1*	3-0-0	3

SSE	*Specialization stream -3*	3-0-0	3	ME	*Program Elective – 2*	3-0-0	3
SSE	*Specialization stream -4*	3-0-0	3	ME	*Program Elective - 3*	3-0-0	3
NG	HSN7XX0: Technical Communication	1-0-0	1	NG	HSN7XX0: Professional Ethics	1-0-0	1
Total (Graded + Non-graded)			**16**	*Total (Graded + Non-graded)*			**15**
III Semester				**IV Semester**			
MP	Project-1	0-0-10	5	MP	Project-2	0-0-22	11
ME	*Program Elective - 4*	3-0-0	3	NG	OAL7XX0: IP Management and Exploitation	1-0-0	1
ME	*Program Elective - 5*	3-0-0	3				
ME	*Program Elective - 6*	2-0-0	2				
MO	*Open Elective-1*	3-0-0	3				
NG	OAN7XX0: Systems Engineering and Project Management	1-0-0	1				
Total (Graded+ Non-graded)			**17**	*Total (Graded + Non-graded)*			**12**

Program Core Course (PC)

S.No.	***Course Title***	***L-T-P***	***Credits***
1	*Introduction to Sustainable Infrastructure Engineering*	*3-0-0*	*3*

List of *Specialization Stream Elective* (SSE) Courses

(Student may choose any 17 credits from the list below in addition to the mandatory program core)

S.No.	***Course Title***	***L-T-P***	***Credits***
1	*Solar Energy and Application*	*3-0-2*	*4*
2	*Wind turbine and Field Design*	3-0-2	4
3	*Offshore Tidal Energy Harvesting*	3-0-0	3
4	*Offshore Wind Energy Harvesting*	3-0-0	3
5	*Energy Storage*	3-0-2	4
6	*Power Plant Engineering*	3-0-2	4
7	*Smart Grid*	3-0-2	4
8	*Biomass Energy Conversion*	3-0-2	4
9	*Hydrogen and Methanol Economy*	3-0-0	3
10	*Plugin Electric Vehicles*	3-0-2	4
11	*Transmission Infrastructure*	3-0-0	3
12	*Fuel Cell Technology*	3-0-0	3
13	*Electrochemical Conversion and Storage*	3-0-0	3
14	*Advanced Waste Water Treatment*	3-0-2	4
15	*Selected Topics in Distributed Generations*	3-0-0	3
16	*Materials for Electrochemical energy Conversion and Storage*	2-0-0	2
17	*Energy Economics and policy*	1-0-0	1
18	*Environmental Aspects of Power Plant*	1-0-0	1
19	*Hydropower*	3-0-0	3

List of *Elective* (*ME*) Courses

Offering Department	***S. No.***	***Course Title***	***L-T-P***	***Credits***
Electrical Engineering	*1*	*Substation Infrastructure*	2-0-0	*2*
	2	*Gas Insulated Substations*	2-0-0	2
	3	*Power System Protection*	3-0-0	3
	4	*Cyber-Physical Security in Smart Grid*	3-0-0	3
Civil & Infrastructure Engineering	*5*	*Stability of Slopes*	3-0-0	3
	6	*Rock Mechanics and Rock Engineering*	3-0-0	3
	7	*Reliability Engineering*	3-0-0	3
	8	*Structural Dynamics*	3-0-0	3
	9	*Wind and Earthquake Engineering*	3-0-0	3
	10	*Offshore Geo-technology*	3-0-0	3
	11	*Climate Change and Impact*	3-0-0	3
	12	*Air Pollution and Control*	3-0-0	3
	13	*Green Building and Sustainable Materials*	3-0-0	3
	14	*Traffic Flow Theory*	3-0-0	3
	15	*Traffic System and Management*	3-0-0	3
	16	*Soil Stabilization and Ground Improvement*	3-0-0	3
	17	*Foundation Engineering*	3-0-0	3
Mechanical Engineering	*18*	*Advanced Mechanics of Solids*	3-0-0	3
	19	*Theory of Vibration*	3-0-0	3
	20	*Remote Sensing and GIS for Environmental Management*	3-0-2	4
	21	*Solid Waste Management*	3-0-0	3
Chemical Engineering	*22*	*Ocean Thermal Energy Conversion*	1-0-0	1
	23	*Wave Energy Conversion*	1-0-0	1
	24	*Process Safety and Hazard*	3-0-0	3
	25	*Advanced Energy Systems*	3-0-0	3
	26	*Microfluidic Fuel Cell*	1-0-0	1
Physics	*27*	*Optics for Solar Energy Applications*	3-0-0	3
	28	*Energy Harvesting*	3-0-0	3
Computer Science and Engineering	*29*	*Artificial Intelligence – I*	3-0-0	3
	30	*Machine Learning - I*	3-0-0	3
TBD[*]	*31*	*Selected Topic in Energy Science and Engineering*	3-0-0	3

[*] *Topic and course contents focusing on the emerging areas and energy technologies would be designed by the course instructor (or Department)*

Topic Clouds Mapped with *Specialization Stream Elective* Courses (SSE)

S.No.	Courses	Offering Department	Topic clouds
1	Solar Energy and Application	Mechanical Engineering	*Solar radiations; Measurement; Concentrated Solar Power; Energy conversion; Thermal comfort; Passive heating and cooling; Daylighting; Energy efficiency; Evaporative cooling; Zero energy building; Building modeling and simulation; Solar PV; Grid integration; MPPT techniques; Solar panels*
2	Wind Turbine and Field Design	Civil & Infrastructure Engineering	*Wind Measurement and statistical distribution; 2D and 3D Aerodynamics; Anatomy of wind turbines; Design of rotor blades; Horizontal-axis wind turbine; Vertical-axis wind turbine; Aeroelasticity; Control mechanisms; Pitch and stall regulation*
3	Offshore Tidal Energy Harvesting	Civil & Infrastructure Engineering	*Aerodynamic and hydrodynamic loads; Rotor and mooring dynamics; Tidal dynamics and ocean currents; Tidal energy system design; Control and stability of tidal power plants; Failure models*
4	Offshore Wind Energy Harvesting	Civil & Infrastructure Engineering	*Offshore wind; wave environment; Fixed and floating offshore platforms; Wind load on structures; Coupled & uncoupled analysis; Fatigue analysis; Offshore wind turbine field design; Vessel-fleet optimization; Cost-benefit evaluation; Power Ratio*
5	Power Plant Engineering	Mechanical Engineering	*Thermal power plant; Nuclear power plant; Hydroelectric power plant; Solar thermal power plant; Hybrid power plant; Plant layout, design and Infrastructure; Boilers; Condensers; Gas turbine; Wind turbines; Fuel cell; Plant load and efficiency; Tidal energy; Geothermal energy; Generator; Transformer; Energy cost*
6	Energy Storage	Electrical Engineering + Mechanical Engineering	*Energy storage; Thermal energy storage; Sensible and latent heat storage; Electrical energy storage; Capacitors and supercapacitors; Chemical and Electrochemical energy storage; Electrolysis; Batteries; Hydrogen storage; Mechanical energy storage; Flywheel; Pumped Hydro; Compressed air energy storage; Storage Applications;*
7	Smart Grid	Electrical Engineering	*Smart Grid Basics; Energy Resources;* *Plug-in Electric Vehicle (PEV); Demand-side management; Monitoring and Protection*
8	Biomass Energy Conversion	Chemical and Engineering + Bioscience and Bioengineering	*Biofuels and bio power; Biomass characteristics; Pre-treatment and processing; Gasification; Pyrolysis; Anaerobic fermentation; Biogas*

			production; Bioreactors; Process kinetics; Algal biofuels; Microbial fuel cell; Hydrogen production; Artificial photosynthesis
9	Hydrogen and Methanol Economy	Chemical Engineering	*Hydrogen and methanol as fuels; Economic, Environmental, Sustainability aspects; Production of hydrogen and methanol as fuel; Challenges and opportunities; Infrastructure and logistics of Hydrogen and methanol production*
10	Plugin Electric Vehicles	Electrical Engineering	*Introduction to EV; Powertrain Storage systems; Charging Infrastructure; Vehicle-to-X systems; Hybrid Electric Vehicle; Vehicle Communication; PEV integration*
11	Transmission Infrastructure	Electrical Engineering	*Overhead transmission lines; Underground cables; Marine cables*
12	Environmental Aspects of Power Plant	Civil & Infrastructure Engineering	*Types of pollutants; Emission standards; Gasification; Combustion; Particulate matter control; Flue gas desulfurization; Types of ash; ash handling; Sustainable practices*
13	Fuel Cell Technology	Chemical Engineering	*Basic principle of Fuel cell technology; Kinetics of Fuel cell electrodes; Charge and mass transport; Modeling and characterization of fuel cell; Principles of proton exchange membrane*
14	Electrochemical Conversion and Storage	Chemical Engineering	*Introduction to electrochemical Conversion and Storage; Introduction to battery and Electrochemical supercapacitor; Principles of electrodialysis; Electrochemical mechanism of Hydrogen production*
15	Advanced Waste Water Treatment	Chemical Engineering	*Introduction to Waste water engineering; Principles of Conventional Waste Water Treatment; Aerobic and Anaerobic processes; Mechanism of Advanced Treatment; Different unit operations for water recycling and reuse*
16	Selected Topics in Distributed Generations	Electrical Engineering	*Distributed generation introduction; Control of distributed generation Microgrids; Stability Challenges in grid integration; Ancillary services*
17	Hydropower	Mechanical Engineering + Electrical Engineering + Civil & Infrastructure Engineering	*Large scale hydropower; Hydropower turbines; Tidal power; Hydropower generators; Small Hydropower; Pumped storage hydropower; Hydropower house; Site investigation; Design of embankment dams; Concrete face rockfill dams; Concrete gravity dams; Foundation preparation*
18	Materials for Electrochemical energy Conversion and Storage	Metallurgical & Materials Engineering + Physics	*Electrochemical energy storage; Conversion devices; Types of batteries; Battery capacity and life; Materials for batteries; 2D and 3D batteries; Principle and types of fuel*

			cells; Performance of fuel cells; Types of solid oxide fuel cells; Hydrogen storage; capacitor and supercapacitors; Challenges and applications
19	Energy Economics and policy	Humanities & Social Sciences	*Energy resources; Energy accounting; Demand analysis; Demand and load management*

Course Content of *Program Core* Course (PC)

Title	**Introduction to Sustainable Infrastructure Engineering**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-Th [C]	3-0-0 [3]
Offered for	M.Tech/M. Tech-PhD	Type	Compulsory
Prerequisite	B.Tech Degree		

Objectives: This course aims at

1. Presenting elements and components of sustainable infrastructure creation and management.
2. Introducing practices of water management, greenhouse gas reduction, and green city planning.

Learning Outcomes

The students will acquire-

1. Fundamentals and principles of sustainable engineering design.
2. Knowledge of existing frameworks to achieve long-term sustainability.
3. Methods and techniques for waste water management and greenhouse gas emission reduction.
4. Knowledge on the engineering and architectural aspects overall green city planning.

Content:

- *Process of sustainable engineering design:* key principle of sustainability; Integration of architecture and engineering; design drivers for sustainable infrastructure system; implementation.
- *Sustainable infrastructure framework:* Green building rating system; sustainable infrastructure frameworks: pillars of sustainability, scale-density framework, transect, built form-ecology framework.
- *Water conservation and supply:* water management plans; achieving water balance; analyzing water resources; water supply strategies.
- *Integrated water management:* integrated stormwater management; urban stormwater treatment strategies; constraints and barriers to implementation; graywater treatment and reuse; integrating graywater into a water resources master plan; blackwater management approaches.
- *Energy and greenhouse gases:* reducing demand by design; designing sustainable power supplies; addressing climate change and reducing carbon footprint; policy measures for increasing energy security and efficiency; design guidelines and performance standards.
- *Sustainable site planning, built systems, and material flows:* sustainable site planning; green streets and transportation network; working with the land; material and waste flows.
- *City*-scale *approach:* Case study1 (Guangzhou: city-scale transformation in China); Case study 2 (Tianjin: Eco-city master plan); Case-study 3 (PlaNYC: integrated stormwater approach); Case study 4 (San Francisco city greening initiative)
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Reference Books

1. S. Bry Sarté (2010) Sustainable Infrastructure. John Wiles & Sons, Inc. ISBN 978-0-470-91293-5.

Learning Material

Detailed Course Content of Specialization Stream Courses (SSE)

Title	**Biomass Energy Conversion**	Course	BBXX
Department	Bioscience and Bioengineering	L-T-P [C]	3-0-2 [4]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives

The Instructor will:

1. Illustrate the principles and applications of bio-energy harnessing systems ranging from conventional processes to the latest developments in the area.

Learning Outcomes

The students are expected to have the ability to:

1. Have an understanding of the existing and emerging biomass to energy technologies.
2. Utilize the knowledge of bioenergy systems while integrating bioenergy with other renewable energy systems.
3. Suggest potential solutions for energy needs by incorporating bioenergy technology.

Contents

Bioenergy context: Introduction, Biofuels and biopower; automobile and biofuels, feedstocks (harvested and residual), Emissions from biofuels *[Lecture 6].*

Thermochemical biomass conversion: Chemical composition, Biomass types and characteristics, Biomass pre-treatment and processing, Combustion, Gasification, Pyrolysis, Catalytic conversion of biomass to liquid biofuels, Life cycle analysis *[Lecture 14].*

Biochemical biomass conversion: Anaerobic fermentation technology: Biogas production, process microbiology, different types of bioreactors, basic process kinetics, Various types of biofuels,1st and 2nd generation biofuels, Algal Biofuels, ASTM standards for biofuels, Engineered microbes for biofuel *[Lecture 10].*

Bioelectrochemical systems: Introduction to Microbial Fuel cells (MFC), Nernst equation and electrochemical potential difference, the redox reactions, calculating and measuring potentials, The exoelectrogenic microbes, MFC components, Applications of MFC, Bio electrochemical system for Hydrogen production, Artificial photosynthesis *[Lecture 12].*

Lab: Biomass acid treatment process and estimation of reducing sugars, performing ethanol fermentation and determine ethanol yield and productivity, performing butanol fermentation, setting up a biogas production process, Screening cellulose producers from natural environments, enzyme activity assays for lignocellulosic waste, performing acid catalysed/base catalysed biodiesel production-comparison of process kinetics, determine biodiesel yield with sequential base/acid treatment for free fatty acid rich microbial oil, Setting up Microbial fuel cells,

Textbooks

1. German Solar Energy Society, Ecofys, (2006) *Planning and installing bioenergy systems: A guide for installers, architects and engineers*, Earthscan,
2. Vertes, A. A., Qureshi, A., Yukawa, H., Blaschek, H.P.,(2010) *Biomass to Biofuels,* Wiley.
3. H. S. Mukunda, *Understanding Clean Energy and Fuels from Biomass*, Wiley India Pvt. Ltd., 2011.

Reference Books

1. Stanier, R. Y., (2009) *General Microbiology*, W.H. Freeman and Co.
2. Pelczar, M.J., Chan, E. C. S., Krieg, N.R.,(2001) *Microbiology*, Tata McGraw - Hill Education

Online course material

Michelle O'Malley, *Fueling Sustainability: Engineering Microbial Systems for Biofuel Production,* Spring 2011, MIT OpenCourseWare Massachusetts Institute of Technology, https://ocw.mit.edu. License: Creative Commons BY-NC-SA

Title	**Solar Energy and Applications**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-2 [4]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite	B.Tech Degree		

Objectives

This course has objectives-

1. to familiarize students with the potential of solar energy to meet the growing energy demands of society in a sustainable way,
2. to elaborate students, the concepts of solar energy applications in architecture and infrastructure, and
3. to illustrate solar technologies of producing power and cooling/heating in buildings.

Learning Outcomes

The students will acquire-

1. knowledge of various means of solar energy utilization technologies and their applications.
2. ability to model, design and evaluate solar energy systems and their applications from the fundamental principles and methods.
3. practical exposer to various measurement experiments and simulation of applications.

Content:

Solar Radiation: Introduction to Solar Radiation, distribution, measurement and data, Sun motion, Introduction to Detectors, Radiation on Sloped Surfaces, Principle of conversion of solar radiation into heat, Thermal comfort, Introduction to Concentrated Solar Power (CSP) Technologies. *[Lectures 14]*

Photovoltaics: Photovoltaic (PV) Effect, PV cells, The electrical model of PV cells, effect of shading, converters for grid integration of solar PV: DC-DC boost converter and inverter, control of boost converter and inverter, MPPT techniques and implementation. Mounting of solar panels, design and analysis of solar trees. *[Lectures 14]*

Energy Efficient Buildings: Building orientation and design, passive heating and cooling concepts, thumb rules, heat transfer in buildings: Thermal modeling of passive concepts, Evaporative cooling, Energy efficient windows and daylighting, Earth air tunnel and heat exchanger, Zero energy building concept and rating systems, Energy conservation building codes, Software for Building Simulation, Automation and Energy Management of Buildings. *[Lectures 14]*

Laboratory Classes [12-13 Classes]

Experiments: Measurement of solar radiation, Measurement of-emissivity, reflectivity, transmittivity, Performance testing of forced flow & thermosyphon systems, Performance testing of- solar air heater, dryer, desalination unit, solar flat plate collector, solar concentrators, Testing and characterization solar PV.
Computational hands-on: Simulation of- Heating and cooling of buildings, Hot water storage system, Heat exchangers, Grid integration of Solar PV, etc.

Reference Books

1. Solar Engineering of Thermal Processes. John A. Duffie, William A. Beckman. John, Fourth Edition, Wiley & Sons, Inc., 2013.
2. Renewable Energy Resources: Basic Principles and Applications. G. N. Tiwari, M K. Ghosal, Narosa Publishing House, New Delhi, 2004.
3. M.S. Sodha, N.K. Bansal, P.K. Bansal, A. Kumar, and M.A.S.Malik, Solar Passive Building, Science and Design, Pergamon Press, 1986.
4. Masters,G.M., (2004), *Renewable and Efficient Electric Power Systems*, John Wiley & Sons, Inc.

Learning Material

4. Solar Energy Technology https://nptel.ac.in/courses/112/105/112105051/

5. Solar Photovoltaics Fundamentals, Technology And Applications https://nptel.ac.in/courses/115/107/115107116/

Title	**Wind turbine and field design**	Number	**CIE7XX0**
Department	Civil Engineering	L-T-P [C]	3–0–2 [4]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives

1. To provide high-level insight into the technical and engineering aspects of wind energy.
2. To develop understanding of advanced principles of aerodynamics and structural mechanics used in design of wind turbine.

Learning Outcomes

The students will have understanding of:

1. Wind turbine dynamic system behavior with integrated components and their control for a given application.
2. Computing performance characteristics of the main components of wind turbine.
3. Wind turbine blade aerodynamics to predict force components and moments experienced by modern wind turbines.

Contents

Wind power: Introduction to wind power: Maximum rated power, power coefficient, average annual power from a wind turbine, wind speed relationships based on the Beaufort scale, wind turbine protection from the wind, total energy from the wind, energy content, geographical resource distribution, influence of terrain, measurement methods and statistical analysis. (Lectures: 10)

Aerodynamics: Fluid flow concept, 2-D Aerodynamics, 3-D aerodynamics. (Lectures: 12)

Turbines: Introduction and Evolution of wind turbines, classification and status and basic anatomy of wind turbines, design of turbine rotor blades, HAWTs, VAWTs, Betz and Glaurets turbine theories and the BEM method, Horizontal and vertical axis wind turbines (HAWTs and VAWTs), blades, choice of materials and adaptation to different climates, aeroelasticity and Wind turbine modelling. (Lectures: 10)

Controls: Control mechanisms, control targets, control strategies (pitch and stall regulation). (Lectures: 10)

Laboratory: Modelling of (a) drivetrain (b) stress generation on wind blades (c) Flow Modelling Around a Wind Turbine; (d) neck and welding points stress modelling

Textbook

1. J. F. Maxwell, J. G. McGowan, and A. L. Rogers, Wind Energy Explained – Theory, Design, and Applications, John Wiley & Sons, 2010, ISBN: 978-0-47001500-1.
2. Martin O. L. Hansen, Aerodynamics of Wind Turbines, Second edition, Earthscan, UK, 2008.

Title	**Offshore Tidal Energy Harvesting**	Number	CIE7XX0
Department	Civil Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives

1. Holistic introduction of different aspects of a tidal energy system.
2. Introduction of design principles and size optimization.
3. Basics of control and reliability assessment of a tidal system.

Learning Outcomes

1. Knowledge of prefeasibility assessment models.
2. Knowledge of optimization techniques to increase efficiency of a tidal energy system.
3. Different control system of a tidal current turbine.
4. Knowledge of different failure models for reliability assessment.

Content

Introduction of tidal energy: Principles and components of tidal power plants; Available technologies and concepts; Estimation of energy calculation; Tidal dynamics and structures of tidal currents. *(Lectures: 6)*

Prefeasibility analysis: Analysis of location; Physical boundaries of assessment; Static v/s transect field survey; Location assessment: Farm method and Flux method. *(Lectures: 6)*

Optimum sizing and modelling of tidal energy system: Modelling of tidal energy conversion systems; Numerical solution of tidal energy systems; Modelling of a tidal current turbine; Tidal energy facility size. *(Lectures: 6)*

Hydrokinetic turbine: Basic principles; emerging designs: Unducted horizontal-axis turbines, Ducted horizontal-axis turbines, Unducted cross-flow turbines, Ducted cross-flow turbines, Gorlov helical turbine, Vortex-induced-vibration systems, Venturi systems, Oscillating hydrofoils, Paddle wheels, Flutter vane, Fan belt; Mooring systems and power transmission. *(Lectures: 6)*

Control system: Automatic control system; Control strategies of ocean wave conversion systems; Reactive power control of tidal turbine; Dynamic control of tidal current turbine; Power system stability analysis. *(Lectures: 6)*

Reliability assessment: The failure distribution model; Constant failure rate model; Time dependent failure model; Assessment of tidal energy system by fault tree analysis. *(Lectures: 6)*

Advances in tidal energy systems: Tidal energy system scenario: India and World; Environmental assessment; Tidal harmonic analysis; Assessment of tidal energy system: Optimization technique and Game theory; Economic analysis. *(Lectures: 6)*

Textbook

1. Vikas Khare, Cheshta Khare , Savita Nema, Prashant Baredar (2019). Tidal energy systems. Elsevier. ISBN: 978-0-12-814881-5.
2. S. K. Chakrabarti (2021). Dynamics of floating offshore structures. World Scientific. ISBN: 9789814280563

Other references:

1. Lago LI, Ponta FL, Chen L (2010) Advances and trends in hydrokinetic turbine systems. Energy for Sustainable Development. Volume 14, Issue 4, Pages 287-296. https://doi.org/10.1016/j.esd.2010.09.004.

Title	**Offshore Wind Energy Harvesting**	Number	**CIE7XX0**
Department	Civil Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M.Tech, M.Tech.-Ph.D. [CIE]	Type	Elective
Prerequisite			

Objectives

1. To highlight the complexities of an offshore wind farms and differences with their onshore counterpart.
2. To introduce the principles of an offshore wind farm design.
3. To introduce principles of management and economics of a wind farm

Learning Outcomes

The students will have understanding of:

1. Aerodynamic, hydrodynamic, and structural loads on offshore wind structures.
2. Design of fixed and floating platforms for wind turbines.
3. Principles of designs of a wind firm, its control, and maintenance.
4. Details of the life-cycle cost and economic feasibility study of an offshore wind farm.

Contents

The offshore environment: Atmospheric wind flow and its characteristic in the boundary layer, ocean winds, wave theories, wind and wave forces on offshore structures *(Lectures: 7)*

Fixed and floating offshore platforms, their analysis, and design: Fixed and floating offshore platforms, mooring lines-their dynamic analysis, wind forces on offshore structures, Design provisions in API and DNV codes, Fixed and floating offshore wind turbines-their dynamic analysis, coupled and uncoupled analysis, Fatigue consideration *(Lectures: 8)*

Floating offshore wind turbines (FOWT): OWT drivetrain and gearbox simulation; Dynamic behaviour of bearings on offshore wind turbine gearbox: Experimental characterization of OWT gearbox in operation. *(Lectures: 7)*

Offshore wind turbine field design: Spatiotemporal wind resource assessment; Bathymetry and other sea constraints; Strategic planning; Offshore wind farm design; Large eddy simulation of wind farm aerodynamics. *(Lectures: 7)*

Operation and Maintenance Modelling: Classification of models; State-of-the-art in modelling; Decision support tools: vessel-fleet optimization models, cost-benefit evaluation of remote inspection. *(Lectures: 7)*

Life-cycle cost and economic feasibility of a floating offshore wind farm: Life-cycle cost calculation: Economic Map Tools and Restrictions Maps Tools; Life-cycle definition, breakdown structure of processes, and calculation of costs; CAPEX and OPEX; Net Present Value (NPV); Internal Rate of Return (IRR); Discounted Pay-Back Period (DPBP); Levelized Cost of Energy (LCOE); Cost of Power Ratio. *(Lectures:7)*

Suggested Textbooks:

1. Olimpo Anaya-Lara, John O. Tande, Kjetil Uhlen, Karl Merz (2018) *Offshore Wind Energy Technology*. Wiley. ISBN 9781119097785.
2. Joao Cruz, Mairead Atcheson (2016). *Floating offshore wind energy (1st Ed).* Springer. ISBN 978-3-319-29398-1.
3. Wiesław Ostachowicz, Malcolm McGugan, Jens-Uwe Schroder-Hinrichs, Marcin Luczak. *MARE-WINT. New Materials and Reliability in Offshore Wind Turbine Technology*. Springer Nature. ISBN 978-3-319-39034-9.
4. Laura Castro-Santos, Vincente Diaz-Casas (2016) *Floating Offshore Wind Farms*. Springer. ISBN 978-3-319-27972-5.
5. S. K. Chakrabarti (2021). *Dynamics of floating offshore structures*. World Scientific. ISBN: 9789814280563

Title	**Energy Storage**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P [C]	3–0–2 [4]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives

The Instructor will:

1. Provide understanding of various types of energy storage elements, their principle, characteristics, and applications

Learning Outcomes

The students are expected to have the ability to:

1. Design suitable energy storage technology as per demand, location, cost and other parameters.
2. Test and characterize various batteries/systems under different operating and ambient conditions

Contents

Fractal - 1

Energy Storage Systems: Introduction of energy storage systems, Energy Demand, Energy Storage Methods and comparison. (4 Lectures)

Thermal energy storage systems: Basic Principle, Thermal energy storage methods, Sensible heat storage. (5 Lectures)

Latent heat storage, phase-change materials, operational aspects, Cold thermal energy storage, Seasonal thermal energy storage (5 Lectures)

Fractal - 2

Mechanical energy storage systems: Introduction, Flywheel energy storage-design and components (4 Lectures)

Pumped hydropower storage (3 Lectures)

Compressed-air energy storage- Specifications of components, classification. (4 Lectures)

Applications, Technical characteristics in terms of power rating and discharge time, storage duration, energy efficiency, energy density, cycle life, lifetime, capital cost, etc. (3 Lectures)

Fractal - 3

Electrochemical/chemical energy storage: Introduction, rechargeable batteries and their electrochemistry, types of batteries - integration into electrical grids, performance, characteristics, off-grid systems - sizing - small scale battery storage systems, recycling technologies, battery testing, battery management, regulations and safety aspects of high voltage batteries, hydrogen energy storage Hydrogen economy, fuel cell, safety (11 lectures)

Electrical energy storage: Superconducting magnetic energy storage systems - principles, materials, performance, and characteristics, super capacitors (3 lectures)

Laboratory Classes [12-13 Classes] The experiments for the course include:
Measurement of internal resistance, cold-cranking amps, capacity, state of health, charging efficiency; Charging and discharging characteristics; Testing of stater-batteries; Self-discharge and storage; Effect of temperature on – voltage, capacity, internal resistance, self-discharge, life; Centralized, distributed, hybrid cell balancing; Design and simulation of charge controllers for battery; Simulation of battery management system; Simulation of different models using different software like Siemens Simcenter battery; design studio, Matlab.

Textbook

1. Cabeza,L.F., (2015), *Advances in Thermal Energy Storage Systems: Methods and Applications*, Woodhead Publishing
2. Kalaiselvam,S., Parameshwaram R., (2014), *Thermal Energy Storage for Sustainability-Systems Design, Assessment and Applications*, Academic Press

3. Alami,A.H., (2020), *Mechanical Energy Storage for Renewable and Sustainable Energy Resources*, Springer
4. Huggins,R.A., (2014), *Energy Storage-Fundamentals, Materials and Applications*, 2nd edition, Springer

Self Learning Material

Title	**Power Plant Engineering**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P-Th [C]	3-0-2 [4]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite	B.Tech		

Objectives

This course is designed-

1. to introduce different aspects of power plant engineering such as power plant layout, operation and safety principles
2. to provide knowledge of the working principle of conversion of different energy resources into electrical power generation.

Learning Outcomes

The students will acquire-

1. ability to select an appropriate type of power plant for given requirements under different situations
2. skills for optimization of energy efficiency of power plants with respect to available resources and energy requirements.
3. ability to analyze the working and layout of power plants and the various components comprising the plant.

Content:

Introduction to power plant engineering: Sources of energy, National and International status on power generation. (Lectures: 2)

Thermal Power Plant: Plant layout, Thermal power plant equipment, Types of boilers, the performance of boilers, supercritical steam generator, the performance of condensers, types of steam turbines, draught system. (Lectures: 8)

Gas Turbine Power Plant: Classification and construction layout, Principles of closed and open cycle gas turbines. (Lectures: 4)

Combined cycle power plants: Load type and calculation, Load variation, Use of internal combustion engine plant, Plant layout and design calculation, combined cycle efficiency. (Lectures: 5)

Nuclear power plant: Principles of nuclear energy, Elements of nuclear power plants, Types of nuclear fuel, nuclear reactor, Boiling Water Reactor, Pressurised Water Reactor, Pressurised Heavy Water Reactor, Fast breeder reactors, Next generation nuclear plant. (Lectures: 6)

Hydroelectric power plant: Plant layout, water turbine, plant operation, water storage and flow duration, hydrographs, Low, medium and high head plants, pumped storage plants, Penstock, water hammer, surge tanks, gates and valves, power house. (Lectures: 6)

Site selection: Selection of for power station, load estimation, load duration curve, load factor, capacity factor, use factor, diversity factor, demand factor, Effect of variable load on power plant, selection of the number and size of units. (Lectures: 4)

Economics and Environmental Considerations: Capital cost, Cost of energy production, Operational cost of power plants, tariffs for electrical energy, Environmental impact, different types of pollution, fuel availability. (Lectures: 3)

Electrical components of power plants: generator, transformer, speed governing systems, generator excitation systems, switchyard. (Lectures: 4)

Laboratory Classes [12-13 Classes]

Steam Power Plant: Start-up, Operation and maintenance, Shut down of Steam Power Plant, Control and monitoring.

Boilers: Efficiency of a steam generator, Analysis of exhaust gas from boiler, Determination of the heat flux density and the overall heat transfer coefficient in boilers.

Steam Turbine: Demonstration of the function of a steam turbine, Operating Characteristics of Steam turbine.

Cooling tower: Determination of cooling capacity, Operating Characteristics of Cooling tower

Gas Turbine: Determination of specific fuel consumption, Determination of Shaft power and System efficiency.

Reference Books

1. M. M. Ei-Wakil, Power Plant Technology, Tata McGraw-Hill.
2. P. K. Nag, Power Plant Engineering, Tata McGraw-Hill.
3. F. T. Morse, Power Plant Engineering, D. Van Nostran, New York.

Learning Material

1. https://nptel.ac.in/courses/112107291/

Title	**Smart Grid**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P [C]	3–0–2 [4]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives
The Instructor will:
1. Provide concepts and topics that are relevant to smart grid technologies to facilitate exploring research opportunities

Learning Outcomes
The students are expected to have the ability to:
1. Understand the basic concepts of smart grid development and the critical technologies that underpin such development, their basic principles, physical constraints, and technological potentials

Contents

Smart Grid Basics: Evolution of Electric Power Grid and Smart Grid, Objectives, main features and challenges of smart grid (5 lectures)
Energy Resources: Centralized vs. distributed generation (1 lecture); renewable energy: solar, wind, hydropower, biomass, geothermal, ocean wave; benefit, costs, and policies of renewable energy (5 lectures); renewable sources integration – overcoming intermittence; storage systems technology (4 lectures)
Plug-in Electric Vehicle (PEV): History of EV; PEV challenges and potential solutions (1 lecture); EV and electric power grid; PEV charging infrastructure, challenges and solutions (4 lectures); PEV as an energy storage device and an energy source (V2G) (2 lectures)
Demand-side management: Load profile of the power grid; market pricing (3 lectures); peak shaving and valley filling; load forecasting (4 lectures); regulations and policies (3 lectures)
Monitoring and Protection: Wide-area monitoring system (WAMS), SCADA and PMU (4 lectures); advanced metering infrastructure (AMI); smart metering (3 lectures); communication infrastructure and technologies (3 lectures)

Laboratory: Experiments related to the following topics will be conducted under laboratory component: Characteristics of solar PV, grid integration of renewable energy sources, effect of loads on power quality with renewable energy penetration, Inverter-based microgrid, virtual inertia in inverter-based sources, demand side management, contribution of energy storage systems to enhance power quality, ancillary services, smart metering, performance of PMU

Textbook
1. Bollean,M.H.J., Hasan,F., (2011), *Integration of Distributed Generation in the Power Systems*, Willey-IEEE India Press
2. Ehsani,M., Gao,Y., Gay,S.E., Emadi,A., (2005), *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design*, CRC Press
3. Bansal,R., (2019), *Power System Protection in Smart Grid Environment*, CRC Press

Self Learning Material
1. The Smart Grid: An Introduction,
http://www.oe.energy.gov/SmartGridIntroduction.htm, Department of Energy, 2008.

Title	**Hydrogen and Methanol Economy**	Number	CH7XX0
Department	Chemical Engineering	L-T-P [C]	3-0-0
Offered for	MTech, MTech-PhD, PhD	Type	SSE
Prerequisite	None		

Objectives

The Instructor will:

1. Provide in-depth knowledge about Hydrogen and Methanol production, storage, their Utility and infrastructure development.
2. Impart knowledge about the economic and environmental viewpoint of using Hydrogen and Methanol as fuel sources.

Learning Outcomes

The students are expected to have:

1. Proper knowledge about Hydrogen and Methanol production and related issues.
2. Economic view point of Hydrogen and Methanol as fuel and the related challenges & opportunities.

Contents

Hydrogen and methanol as fuels: General aspects, Prelude to energy systems, Hydrogen and Methanol as energy sources, Future energy supply, Economy and Environment, Sustainability (*Lectures: 6*)

Hydrogen Production Processes: Properties, production, storage, utilization and safely, Infrastructure requirement, Hydrogen as a transportation fuel, Hydrogen based fuel cell. (*Lectures: 12)*

Methanol Production Processes: Properties, production, storage, transportation and safely, Infrastructure Requirement, Methanol as fuel and energy carrier, Methanol as automobile, marine and aviation fuel. (*Lectures: 14*)

Challenges and opportunities: Indian and global perspectives, Domestic and transportation application, Affordable Technology development, Policy. (*Lectures: 6*)

Infrastructure and other issues: Storage and transmission infrastructure, Local distribution, filling stations and building integrated systems, national and international safety standards, socio-economic implications. *(Lectures: 6)*

Textbook

1. G. A. Olah, A. Goeppert, G. K. S. Prakash,(2006), *Beyond Oil and Gas: The Methanol Economy*, Wiley-VCH, Weinheim, Germany.
2. Ball M. and Wietschel M.,(2009) *The Hydrogen Economy Opportunities and Challenges*, Cambridge University Press.

Reference Books

1. Scipioni, A. Manzardo, J. Ren, (2017) *Hydrogen Economy: Supply Chain, Life Cycle Analysis and Energy Transition for Sustainability*, Academic Press.
2. M. Bertau, H. Offermanns, L. Plass, F. Schmidt, H. Wernicke,(2014) *Methanol: The Basic Chemical and Energy Feedstock of the Future*, Springer; 2014
3. Bent Sorensen, 2012, *Hydrogen and Fuel Cells: Emerging Technologies and Applications*, Elsevier, Second Edition.

Title	**Plugin Electric Vehicles**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P[C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives

The Instructor will:

1. Provide understanding of various technologies of plugin electric vehicle and hybrid electric vehicle and their role in smart grid operation

Learning Outcomes

The students are expected to have the ability to:

1. Design powertrain circuit topology and appropriate control scheme for electric vehicle
2. Understand and design EV charging infrastructure

Contents

Introduction to EV: Recent development trends, types of electric vehicles (2 lectures)
Powertrain: Concepts of powertrain sizing in EV and PHEV, starting, braking (6 lectures)
Storage systems: Battery types, parameters, battery modeling, battery management (5 lectures)
Charging Infrastructure: Domestic charging infrastructure, public charging infrastructure, normal charging station, occasional charging station, fast charging station, battery swapping station (5 lectures), types of charging: constant voltage, constant current, pulse charging, inductive, conductive charging (3 lectures)
Vehicle-to-X systems: Vehicle-to-grid, vehicle-to-home (5 lectures), vehicle-to-building, vehicle-to-vehicle (3 lectures)
Hybrid Electric Vehicle: Configuration of HEV (series, parallel, series-parallel & complex), power flow control in HEV configurations, control (6 lectures)
Vehicle Communication: Introduction to in-vehicle communications and controller area network (4 lectures)
PEV integration: Impact on transmission and distribution network (3 lectures)
Laboratory: Experiments related to the following topics will be conducted under laboratory component: SoC estimation, control of electric vehicle drives, regenerative braking, powertrain and vehicle dynamics, EV chargers and their control, vehicle-to-grid technology, vehicle-to-vehicle technology, EV charging from solar PV, performance of aggregator

Laboratory: Experiments related to the following topics will be conducted under laboratory component:
SoC estimation, control of electric vehicle drives, regenerative braking, powertrain and vehicle dynamics, EV chargers and their control, vehicle-to-grid technology, vehicle-to-vehicle technology, EV charging from solar PV, performance of aggregator, Use of simulation tools (MATLAB, SIEMENS EV related softwares) to simulate different subsystems of EV and different scenarios.

Textbook

1. Husain,I., (2011), *Electric and Hybrid Vehicles Design Fundamentals*, 2nd Edition, CRC Press
2. Ehsani,M., Gao,Y., Emadi,A., (2011), *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles Fundamentals, Theory and Design*, 2nd Edition, CRC Press
3. Larminie,J., Lowry,J., (2003), *Electric Vehicle Technology Explained*, Wiley

Self Learning Material

1. Jain,A.K., *Electric Vehicles - Part 1,* NPTEL Course Material, Department of Electrical Engineering, IIT Delhi, https://nptel.ac.in/courses/108/102/108102121/
2. Kumar,P., Majhi,S., *Introduction to Hybrid and Electric Vehicles,* NPTEL Course Material, Department of Electronics and Communication Engineering, IIT Guwahati, https://nptel.ac.in/courses/108/103/108103009/

Title	**Energy Economics and Policy**	Number	HSS7XX0
Department	Humanities and Social Sciences	L-T-P-D [C]	1-0-0 [1]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives

1. To introduce basics of energy economics and demands.

Learning Outcomes

1. Knowledge of energy classification, calculation, and audit.
2. Knowledge of demand analysis and management.

Contents

1. *Energy as economic resources:* classification of energy resources; measurement of energy; energy accounting. (Lectures: 7)
2. *Energy demand:* basic concepts in economics; descriptive analysis of energy demand; decomposition analysis and parametric approach; demand side management; load management. (Lectures: 7)

Text Books

1. Alessandro Rubino Alessandro Sapio Massimo La Scala Michelle Hallack (2020) *Handbook of Energy Economics and Policy (1st Ed).* Elsevier Academic Press. ISBN: 9780128147122.

Online course Material

1. "*Energy Economics and policy"* by Prof. Shyamasree Dasgupta on NTPEL. https://nptel.ac.in/courses/109/106/109106161/

Title	**Materials for Electrochemical Energy Conversion and Storage**	Number	
Department	Physics and Metallurgical & Materials Engineering	L-T-P [C]	2-0-0 [2]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite	-		

Objectives

The Instructor will:

1. Provide the student with an extended knowledge of the basic principles and recent advances in Energy Storage materials and associated technologies for consumer electronics to high-power applications.
2. Provide the students with the role of materials in energy storage technologies.

Learning Outcomes

The students are expected to gain knowledge on:

1. Different modes of energy conversion and storage with respect to scale, cost, energy and power density
2. Structure processing property performance relationship of the energy materials developed over time.

Contents

Introduction: Overview of electrochemical energy storage and conversion devices, Basic terminology, Classification, Applications. **(Offered by Physics)** (Lectures: 2)

Batteries: Introduction to Batteries, primary and secondary batteries, Solid-state and liquid electrolyte batteries, Lead acid batteries, Nickel Cadmium Batteries, Li-ion and Li-Polymer batteries, Intercalation and Conversion Materials, Battery Capacity and Life, Micro-batteries, Thin film batteries, Materials for high-energy density 2D and 3D batteries **(Offered by Physics)** (Lectures: 12)

Fuel cells: Principle and types, performance characteristics and efficiency of fuel cell, Proton exchange fuel cells, Solid oxide fuel cells, Types of solid oxide fuel cells: High and intermediate temperature, Single chamber solid oxide fuel cells. Polymer electrolyte fuel cells, Acid/alkaline fuel cells, phosphoric acid fuel cell, molten carbonate fuel cell, fuel cell stack, hydrogen storage **(Offered by MT)** (Lectures: 10)

Supercapacitors: Difference between Capacitor and Supercapacitor, Electrolyte and Electrode materials, electric double layer capacitors, Challenges and Applications **(Offered by MT)** (Lectures: 4)

Textbook

1. Cheong K. Y., Impellizzeri G and Fraga M. A. (2018) *Emerging Materials for Energy Conversion and Storage*, 1st Edition, Elsevier

Reference Book

1. Rosen M. A. (2012) *Energy Storage*, Nova

Online Course Materials

1. Prof. Bazant M., MITopencourseware: Electrochemical Energy Systems, MIT Course Number 10.626 / 10.426, https://ocw.mit.edu/courses/chemical-engineering/10-626-electrochemical-energy-systems-spring-2014/

2. Prof. Pal K., Selection Of Nanomaterials For Energy Harvesting And Storage Application, NPTEL Course Material, Mechanical Engineering, Indian Institute of Technology Roorkee, https://nptel.ac.in/courses/112/107/112107283/

Title	**Hydropower**	Number	MEL7XX0
Department	Mechanical Engineering, electrical Engineering, Civil & Infrastructure Engineering	L-T-P[C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives:
The Instructor will:

1. Describe the role of the hydropower and working principles of small- and large-scale hydropower plants.
2. Provide overview of the practical design of hydropower structures including mechanical and electrical equipment.
3. Introduce students to geotechnical, mechanical, and electrical engineering aspects of conventional large dams.

Learning Outcomes:
The students are expected to have the ability to:

1. Formulate and analyze technical solutions for design, and operation of hydropower plants.
2. Analyze a technical problems connected to the hydroelectric power industry.
3. Compare the design of dams, mechanical, electric and electronic equipment.
4. Implement design details of embankment dam, components of face rockfill dams and concrete gravity dam, and basis of dam foundation design.

Fractal 1 (ME)

- *Introduction*: Hydropower resource, potential and market, Hydropower sites, Types of hydropower plants. *(Lectures: 2)*
- *Large hydropower*: Run-of-River Hydropower, Dams and Barrages, Dam types *(Lectures: 3)*
- Hydropower turbines, Classification and working principles, Impulse and reaction turbines; Propeller and Kaplan Turbines, Deriaz turbine. *(Lectures: 5)*
- *Tidal power*: Tidal barrage power plant, operation and design, Tidal lagoons, Barrage construction techniques, Turbine speed regulation. *(Lectures: 4)*

Fractal 2 (EE)

- *Hydropower Generators:* Generator poles and rotational speed, orientation, types *(Lectures: 2)*
- *Small Hydropower*: Small, mini, and micro hydropower plant, generators for small hydro power plants *(Lectures: 3)*
- *Electronic Control of Hydropower*: Phase angle control, relay switched on loads, Mark-Space ratio controller *(Lectures: 3)*
- *Pumped Storage Hydropower*: Pumped storage hydropower plant, turbines, motors *(Lectures: 3)*
- *Hydropower House:* Classification, equipment, accessories, layout, components, sizing *(Lectures: 3)*

Fractal 2 (CIE)

- *Site investigation techniques:* Geological environments: rocks and soils; Geological, geotechnical, and geophysical investigation; Trenching and drilling *(Lectures: 2)*
- *Design of embankment dams:* freeboard; Slope protection; Embankment crest details; Dimensioning and tolerances; Conduits through embankments; Flood control structures; Seismic stability analysis of embankments; Liquefaction of dam embankments and foundations. *(Lectures: 4)*
- *Concrete face rockfill dams and Concrete gravity dams:* Selection of dam types; Rockfill zones; Concrete face; Construction aspects; Settlements, and displacements of the face slab, and joints; Leakage; Strength and compressibility of rock foundations; Strength of concrete in the dam; Strength of the concrete-rock contact; Silt and ice load *(Lectures: 4)*

- *Foundation preparation:* General foundation preparation for embankment dams; Cutoff foundation for embankment dams; Width and batter slopes; Slope modification; Foundation preparation for concrete gravity dams. *(Lectures: 4)*

Textbook

- Breeze,P., (2018), *Hydropower*, Academic Press
- Pandey,B., Karki,A., (2017), *Hydroelectric Energy, Renewable Energy and the Environment*, CRC Press
- Robin Fell, Patrick MacGregor, David Stapledon, Graeme Bell (2005) *Geotechnical Engineering of Dams*. A. A. Balkema publishers. ISBN 04 1536 440 x.

Title	**Selected Topics in Distributed Generation**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite			

Objectives

The Instructor will:

1. Provide salient concepts of DG operation and control in both grid connected and autonomous modes
2. Provide exposure to the challenges related to stability, grid integration, and ancillary services in distributed generation

Learning Outcomes

The students are expected to have the ability to:

1. Develop control schemes for distributed generators to operate in grid connected and autonomous mode
2. Understand various challenges and propose solutions related to stability, grid integration and ancillary services in distributed generation

Contents

Distributed generation: Introduction, DG types, concept of inertia and frequency response, flexibility issues, ramping issues (3 lectures)

Control of distributed generation: Modelling and control of distributed energy sources such as solar PV, wind energy, diesel generator (10 lectures)

Microgrids: Concept of microgrid, microgrid benefits, typical structures and configurations of various microgrids (3 lectures), control of converters in DC and AC microgrids (4 lectures), active and reactive power control, voltage and frequency control of inverter-based sources (3 lectures), virtual synchronous generators (2 lectures)

Stability: Stability issues with weak grid integration and in microgrid operation (4 lectures)

Other challenges in grid integration: Power quality requirements, DG support under abnormal grid conditions (3 lectures), issues related to islanding and reverse power flow, network congestion (5 lectures), and protection (2 lectures)

Ancillary services: Ancillary services with DGs, control for ancillary services, virtual power plants (3 lectures)

Textbook

1. Bollean,M.H.J., Hasan,F., (2011), *Integration of Distributed Generation in the Power Systems*, Willey-IEEE India Press

2. Masters,G.M., (2004), *Renewable and Efficient Electric Power Systems*, John Wiley & Sons, Inc.

3. Bevrani,H., Francois,B., Ise,T., (2017), *Microgrid Dynamics and Control*, John Wiley & Sons, Inc.

Title	**Advanced Waste Water Treatment**	Number	CHL7XX0
Department	Chemical Engineering	L-T-P (C)	3-0-2 (4)
Offered for	M.Tech., M.Tech-Ph.D., Ph.D.	Type	SSE
Prerequisite	None		

Objective

The instructor will:

1 Impart comprehensive knowledge of industrial and municipal wastewater.
1. Discuss mechanism and methodologies for waste water treatment.
2. Discuss case studies

Learning Outcomes

The students are expected to have the:

1 Comprehensive knowledge of the wastewater treatment.
1. Knowledge of advanced methodologies such as waste water management, recycle and reuse.

Contents

Introduction: Waste water engineering: an overview, characterisation of waste water and monitoring of industrial and municipal waste water, environmental impacts of waste water constituents. (4 lectures)

Conventional Waste Water Treatment: Existing unit operations and processes, basic philosophy of water treatment plants; physio-chemical treatment methods: (i) Screening, (ii) conventional filtration (iii) coagulation, (iv) flocculation, (v) floatation (vi) Clari-flocculation (vii) sedimentation, (viii) sand filtration, etc. (10 lectures)

Aerobic and Anaerobic Suspended and Attached Growth Waste Water Treatment Processes: Aerated lagoon, activated sludge systems, trickling filter, sequential batch reactor, fluidized bed bioreactors, Up-flow Anaerobic Sludge Blanket (UASB) and hybrid Up-flow Anaerobic Sludge Blanket (UASB) reactors. (8 lectures)

Advanced Treatment Processes: Membrane Filtration, reverse osmosis, ultrafiltration, nanofiltration and electrodialysis; wet air oxidation, adsorption and ion-exchange; Wet-land and root-zone treatment of industrial and municipal wastes; Design of sludge drying beds, thermal and biological processes for sludge and land fillings. Membrane Reactors, New Technologies for Waste Water Disinfection. (12 lectures)

Water Recycling and Reuse: Different unit operations for water recycling depending on end use, recovery of valuables from waste water, zero liquid discharge (ZLD). (8 lectures)

Waste Water Treatment Design Lab (28 hrs)

Clarifier; Aerator; Sand filtration system; Municipal waste water system; Grey water treatment system; Sewage water treatment system; Industrial waste water systems; Pond water treatment.

Text Book

1. Burton F.L., Tchobanoglous G., Stensel H.D., 2017, *Waste Water Engineering Treatment and Reuse,* 4thed.,Tata McGraw-Hill.
2. Arceivala S.J. and Asolekar S.R., 2007, *Wastewater Treatment for Pollution Control and Reuse*, 3rd Ed., Tata McGraw Hill.

Reference Books

1. Henze M., van-Loosdrecht M.C.M., Ekama G.A. and Brdjanovic D., 2008, *Biological Wastewater Treatment: Principles, Modelling and Design*, IWA publishing.
2. Sincero A.P. and Sincero G.A., 1996, *Environmental Engineering – A Design Approach,* Prentice-Hall.

Online Course Material

1. Tiwari M K, *Wastewater treatment and recycling*, NPTEL course material, Department of Civil Engineering, IIT Kharagpur, https://nptel.ac.in/courses/105105178/

Title	**Fuel Cell Technology**	Number	CH7XX0
Department	Chemical Engineering	L-T-P [C]	3-0-0
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite	None		

Objectives

The Instructor will:

1. Provide in-depth knowledge about the fuel cell technology.
2. Discuss the fundamentals of deferent fuel cells.

Learning Outcomes

The students are expected to have:

1. Comprehensive knowledge about fuel cell technology.
2. The ability to design different types of cells.

Contents

Basics of Cell Technology: Principles of fuel cells, Fuel cells systems and staking, applications, performance study, electrochemistry and thermodynamics of fuel cells. [6 lectures]

Fuel cell kinetics: electrode kinetics, catalysis in fuel cells, activation energy, reaction equilibrium, exchange currents and electrocatalysis, design principles of catalyst-electrode, Sabatier principle. [6 lectures]

Charge and mass transport: resistance and voltage loss, electrical driving force, fuel cell electrolytes, diffusive and conductive transport, convective transport. [6 lectures]

Modelling and characterization: Theory and governing equations, basic fuel cell model, 1D model, CFD application in modelling, characterization techniques of fuel cells, physical and chemical methods. [6 lectures]

Proton-exchange membrane (PEM) Fuel Cell: Cell components, material properties and processes; membrane layer, current collector and gas delivery, catalyst action, design issues, high temperature and reverse operation, degradation and lifetime. [9 lectures]

Other Fuel cells: Solid Oxide Fuel Cell(SOFC), Phosphoric acid fuel cell (PAFC), Alkaline fuel cell (AFC), Solid acid fuel cell (SAFC), Molten-carbonate fuel cell (MCFC), Microbial fuel cell, etc., Environmental impact of fuel cells, fuel cell vehicles, building-integrated fuel cells. [9 lectures]

Textbook

1. Andrew L. Dicks, David A. J. Rand, 2018, *Fuel Cell Systems Explained*, Willey, Third Edition.
2. Ryan O'Hayre, Suk-Won Cha, Whitney Colella, Fritz B. Prinz, 2005, *Fuel Cell Fundamentals*, Willey, Third Edition.

Reference Books

1. Bent Sorensen, 2012, *Hydrogen and Fuel Cells: Emerging Technologies and Applications*, Elsevier, Second Edition.
2. Frano Barbir, 2013, *PEM Fuel Cells: Theory and Practice*, Elsevier Inc., Second Edition.
3. Ryan O'Hayre, Suk-Won Cha, Whitney Colella, Fritz B. Prinz, 2009, *Fuel Cell Fundamentals*, Wiley, Second Edition.

Title	**Electrochemical Conversion and Storage**	Number	CH7XX0
Department	Chemical Engineering	L-T-P [C]	3-0-0
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite	None		

Objectives

The Instructor will:

1. Impart the knowledge of electrochemical engineering, Working principle, materials and characterization of electrochemical conversion for energy production.
2. Introduce the mechanism to store energy.

Learning Outcomes

The students are expected to have:

1. Comprehensive knowledge about electrochemical process.
2. Advanced knowledge about energy production via electrochemical process and store it efficiently.

Contents

Introduction: basics of electrochemical Conversions and Storage, devices and Principles, over view of fuel cells, Photo-electrochemical conversion and cells. [5 lectures]

Batteries for energy storage: primary and secondary batteries, limitations of battery performance, Lead Acid Battery, Nickel-Metal Hydride (Ni-MH) Rechargeable Batteries, Lithium-Ion Rechargeable Batteries, Liquid- Redox Rechargeable Batteries. [10 lectures]

Electrochemical supercapacitor: introduction to supercapacitor, types and structures, Transport Properties, electric double-layer capacitor, physical/elcrochemical processes in supercapacitors, diffuse layer, pseudocapacitor. [11 lectures]

Electrodialysis: introduction to electrodialysis and reverse electrodialysis, control parameters, effect of flow structure. [4 lectures]

Electrochemical Hydrogen Production: Introduction to electrolysis, electrolysers and hydrogen production, PEM water electrolysers, electrochemical reforming of alcohol, perspectives of hydrogen electroproduction techniques. [12 lectures]

Textbook

1. Vladimir S. Bagotsky, Alexander M. Skundin, Yurij M. Volfkovich, 2015, *Electrochemical Power Sources: Batteries, Fuel Cells, and Supercapacitors*, Wiley.
2. Jiujun Zhang, Lei Zhang, Hansan Liu, Andy Sun, Ru-Shi Liu, 2011, *Electrochemical Technologies for Energy Storage and Conversion*, Wiley-VCH, First Edition.
3. Allen J. Bard, Larry R. Faulkner, 2001, *Electrochemical Methods: Fundamentals and Applications*, John Wiley & Sons, Second Edition.

Reference Books

1. Keith Scott, 2020, *Electrochemical Methods for Hydrogen Production*, RSC, UK.
2. Stanley W. Angrist, 1982, *Direct Energy Conversion* , Allyn and Bacon Inc.

Title	**Environmental Aspects of Power Plant**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	1-0-0 [1]
Offered for	M.Tech, M.Tech.-Ph.D., Ph.D.	Type	SSE
Prerequisite			

Objectives

1. To provide basic knowledge of combustion and conversion technologies.
2. To make students familiar with emissions, pre- and post-combustion control technologies and waste management techniques.

Learning Outcomes

The students are expected to achieve proficiency in:

1. Identifying the issues and appropriately choose the right pollution control techniques for gas and particle phase emissions.
2. Assessing the efficiency of conventional as well as emerging combustion technologies targeting pollution reduction.

Contents

Introduction: Types of emissions; types of pollutants (SOx, NOx, trace elements and other Coal Combustion Products) and their impacts; Emission standards *(Lectures: 2)*

Combustion Technologies: Gasification; Types of gasifier; Gasification vs Combustion; Fluidized bed combustion (FBC); Concept of Integrated Gasification Combined Cycle (ICC); IGCC vs pulverized coal combustion (PCC) *(Lectures: 5)*

Pre- and Post-combustion control: Flue gas desulfurization (FGD); NOx and Sox treatments (low NOx burners, flue gas recirculation and SCR); Particulate matter control systems (Electrostatic precipitators, mechanical collectors, and other devices (particulate scrubbers, baghouse filters, fabric filters etc.)) *(Lectures: 5)*

Ash handling: Ash (fly ash, bottom ash, pond ash, FBC ash, boiler slag, FGD ash); Scrubber slurry; Ash handling and sustainability practices, leaching aspects. *(Lectures: 2)*

Text Books

1. Fuels and Fuel Technology, F. Peter, A. Wheatan & Co. Ltd., First Edition, 1965.
2. Fuels and Combustion, S. Sarkar, Orient Longman, Second Edition, 1990.
3. Hydrogen and Syngas Production and Purification Technologies, Ke Liu, C. Song, and V. Subramani, John Wiley & Sons, 2010.
4. The Chemistry and Technology of Coal, J. G. Speight, CRC Press, 2013.

Online course Material

Title	**Transmission Infrastructure**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	SSE
Prerequisite	Smart Grid		

Objectives

The Instructor will:

1. Provide knowledge related to infrastructure in various modes of power transmission such as underground, marine, and overhead.
2. Provide knowledge related to design, installation, and maintenance of the transmission infrastructure.

Learning Outcomes

The students are expected to have the ability to:

1. Select appropriate transmission infrastructure needed for transmission of electric power based on various factors.
2. Design, install, and maintain power transmission infrastructure.

Contents

Overhead transmission lines: Overhead line routing, aerial survey, ground survey, overhead line foundations, soil investigation, foundation types, foundation design, support structures, steel lattice towers, tower testing, overhead line fittings, suspension clamps, sag adjusters, bi-metal connectors, conductor loads, lightning, surge diverters, insulator arcing horn co-ordination, aerodynamic phenomena, transmission voltage level clearances, wooden poles, distribution voltage clearances, overhead line maintenance, environmental considerations (18 lectures)

Underground cables: Types of cable, insulation and conductor materials, air cable ampacity, underground cable ampacity, short circuit and its effect, thermal & electromechanical stresses, cable termination and joints, pulling the cable, installing cable racks, conduits, cable supports, cable tray and trenches, cable testing, steps for fault location, fault location methods (12 lectures)

Marine cables: Introduction, marine survey, cable routing, cable system design parameters, cable design parameters, cable accessories, cable installation, cable materials & accessories, cable joints and terminations, inspections and testing, maintenance and failure investigation (12 lectures)

Textbook

1. Kiessling,F., Nefzger,P., Nolasco,J.F., Kaintzyk,U., (2003), *Overhead Power Lines Planning Design Construction*, 4th Edition, Springer
2. Ganguli,S.K., Kohli,V., (2016), *Power Cable Technology*, CRC Press
3. Worzyk,T., (2009), *Submarine Power Cables Design, Installation, Repair, Environmental Aspects*, Springer
4. Bayliss,C., Hardy,B., (2012), *Transmission and Distribution Electrical Engineering*, 4th Edition, Newnes publications

Self Learning Material

Detailed Course Content of M. Tech Elective Courses (ME)

Title	**Power System Protection**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

The Instructor will provide knowledge of:

1. Various protection schemes and their applications in power industry
2. Relay coordination, instrument transformers
3. Numerical protection schemes based on signal processing techniques

Learning Outcomes

The students are expected to have the ability to:

1. Design/propose protection schemes for various power system components based on digital signal processing
2. Design relay coordination systems in distribution and transmission

Contents

Overview of Power System Protection: Architecture of protection system, evolution of relays, zones of protections, concept of primary and back-up protection (2 lectures)
Overcurrent Protection: Principles of fuse and overcurrent protection and application to feeder and motor protection, relay coordination in distribution system (6 lectures)
Distance Protection: Principles of distance relaying (2 lectures) and application to transmission system protection, relay coordination (5 lectures)
Differential Protection: Principles of differential protection (2 lectures) and application to transformer, bus bar and generator armature winding protection (5 lectures)
Introduction to Digital Relaying: Sampling, concept of moving window, signal conditioning, signal processing (4 lectures)
Instrument Transformer: Current and voltage transformers in power system protection (6 lectures)
Circuit breakers: Arc initiation and quenching, oil circuit breakers, air blast circuit breakers vacuum circuit breakers, SF6 circuit breakers (5 lectures), current chopping, reverse recovery voltage (3 lectures)
Lightning arresters: Principle of operation and types (2 lectures)

Textbook

1. Singh, L.P., (2017), *Digital Protection: Protective Relaying from Electromechanical to Microprocessor*, 2nd Edition, New Age International Publications
2. Ram,B., Vishwakarma,D.N., (2017), *Power System Protection and Switchgear*, 2nd Edition, Tata McGraw Hill Publications.
3. Phadke,A.G., Thorp,J.S., (2009), *Computer relaying for Power System*, 2nd Edition, John Wiley & Sons Ltd. Publications

Self Learning Material

1. Soman,S.A., *Power System Protection,* NPTEL Course Material, Department of Electrical Engineering, Indian Institute of Technology Bombay, https://nptel.ac.in/courses/108/101/108101039/

Title	**Cyber-Physical Security in Smart Grid**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

The Instructor will:

1. Provide exposure to the various layers of smart grid architecture and security issues related to vital components of smart grid

Learning Outcomes

The students are expected to have the ability to:

1. Understand various cyber layers of smart grid and the threats to the physical components, and design a secure system

Contents

Introduction: Physical components of a grid, communication in smart grid, self-monitoring, self-healing, vulnerabilities of smart grid, cyber-physical security (5 lectures)

Security Architecture of Smart Grids: Security architecture of smart Grid and its layers, typical SCADA network levels, conceptual control system with layering, robust and resilient control, secure network routing, management of information security, user patching (6 lectures)

Advanced Metering Infrastructure and Security Issues: Smart meters, customer gateway, AMI communication network, AMI head-end, AMI security threats, interactions, malicious code planted in the system, denial of service, insider attacks, unauthorized access (4 lectures)

SCADA System Security: Introduction to SCADA, Security Issues in SCADA, Public Information Availability, Platform Vulnerabilities, Network Configuration Vulnerabilities, Network Perimeter Vulnerabilities, DNP3 Threats (5 lectures)

Plug-in Electric Vehicle (PEV) Security Issues: Introduction to PEV Security, Privacy of movement, Secure payment, Critical Infrastructure and Physical Security, Communication issues in PEVs (4 lectures)

Smart Grid Domain Security: IEEE standards for smart grid security, IEEE 802.11, IEEE 802.15.4, IEEE 802.16, Bluetooth and BLE/BT5.0, ZigBee, Z-Wave, Gateway Component, FMEA, STAMP, HAZOP, Smart Grid Information Security (SGIS), SGIS Risk Impact Level (SGIS-RIL) (6 lectures)

Blockchain for Smart Grid: Blockchain: Smart contract, applications of blockchain technology to energy; Energy trading: Transition towards blockchain-based electricity trading market, Internet of Energy (IoE); Distributed Energy Resources: Decentralized operation of interdependent power and energy networks based on Blockchain and security, Electric vehicles: Blockchain for decentralized optimization of energy sources: EV charging coordination via blockchain-based charging power quota trading, Microgrid: microgrid monitoring and control using blockchain. (12 lectures)

Textbook

1. Hossain,E., Han,Z., Poor,H.V., (2012), *Smart Grid Communications and Networking*, Cambridge University Press
2. Berger,L.T., Iniewski,K., (2012), *Smart Grid Applications, Communications, and Security*, John Wiley and Sons Publications
3. Shafie-khan,M., (2020), Blockchain-based Smart Grids, Academic Press

Self Learning Material

Title	**Energy Harvesting**	Number	PHL6XX0
Department	Physics	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

The Instructor will:

1. Provide the student with an extended knowledge of the basic principles and recent advances in Energy harvesting technologies for low-power applications including self-powered or autonomous systems.
2. Provide the students with the role of materials in energy harvesting technologies with a focus on miniaturized and microfabricated devices.

Learning Outcomes

The students will:

1. Develop an understanding of basic principle, concept, working mechanism and applications of various energy harvesting devices.
2. Develop an understanding to design new energy harvesting systems.

Contents

Micro Photovoltaic Module Energy Harvesting: *p-n* Junction and Crystalline Si Solar Cells, Amorphous Silicon Solar Cell, CIGS and CdTe Solar Cell Development, Polymer Solar Cell, Dye-Sensitized Solar Cells (DSSC), Perovskite Solar Cells, Monolithically Integration of Solar Cells with IC, Low-Power Micro Photovoltaic Systems, Maximum Power Point Tracking (*Lectures : 14*)

Mechanical Energy Harvesting: Energy Harvesting from Macro to Micro to Nano, Piezoelectric MEMS Energy Harvesters, Nano-Piezoelectric Generator, Piezoelectric Cantilever Generator, Piezoelectric Transducers. New-generation Triboelectric Nanogenerators. (*Lectures: 8*)

Thermal Energy Harvesting: Overview of Nanoscale Heat Conduction and the Seebeck Effect, Thermoelectric energy harvesting and Power Generation, Thermoelectric figures of merit, Novel nanostructured Thermoelectric materials and devices, Thermal energy harvesting using pyroelectric materials, Thermodynamic cycles for Pyroelectric energy harvesting, Nanostructured and micro-scale materials and devices. (*Lectures: 14*)

Textbook

1. Briand, D., Yeatman, E., Roundy, S., *Micro Energy Harvesting* Wiley, 2015.
2. Shashank, P., Inman, D. J., *Energy Harvesting Technologies*, Springer, 2009.

Self-Learning Material

1. Elvin, N., Erturk, A., *Advances in Energy Harvesting Methods*, Springer, 2013.
2. Jaffe, R., and Taylor, W., *Physics of Energy*, MIT open course, https://ocw.mit.edu/courses/physics/8-21-the-physics-of-energy-fall-2009/
3.

Preparatory Course Material

4. Pal, K., Selection of Nanomaterials for Energy harvesting and Energy Storage Applications, https://nptel.ac.in/courses/112107283/

Title	**Ocean Thermal Energy Conversion Process**	Number	CH7XX0
Department	Chemical Engineering	L-T-P [C]	1-0-0
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite	Thermodynamics		

Objectives

The Instructor will:

1. Provide understanding of Ocean Thermal Energy Conversion (OTEC) cycles, hybrid OTEC systems.

Learning Outcomes

The students are expected to have the:

1. Proper knowledge about Ocean Thermal Energy Conversion (OTEC) cycles and hybrid OTEC technologies.
2. Ability to understand OTEC process design challenges.

Contents

Ocean as a Heat Engine: Concepts of ocean thermal energy, Favourable temperature difference, OTEC history. (2 lectures)

OTEC Cycles: Closed OTEC cycles, Open OTEC cycle, Flow diagrams, Comparison of OTEC Cycles, Products of OTEC: electricity, desalinated water, cold water for air-conditioning etc., Hybrid OTEC systems. (6 lectures)

OTEC Process Design: System components, Heat exchangers, evaporators, Pumps, Turbines, Piping. (6 lectures)

Textbook

1. Takahashi, Patrick and Trenka, Andrew, (1996) *Ocean Thermal Energy Conversion,* Ist ed., Wiley.

Reference Books

1. Lockerby,R.W., (1981) *Ocean Thermal Energy Conversion (OTEC),* US Department of Energy, United States.

Title	**Optics for Solar Energy Applications**	Number	PHL6XX0
Department	Physics	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

The students will gain

1. Knowledge to understand the issues related to optics in solar thermal applications.
2. Knowledge in designing and analyzing optics for imaging and non-imaging optics.

Learning Outcomes

The students are expected to have the ability to:

1. Show a basic familiarity with the Concentrated Solar Power (CSP) plants.
2. To design as individual optical concentrators and Optical Field Design.

Contents

Basics of Maxwell equation: Review of Electromagnetic theory (2 Lectures)
Wave Optics: Polarization (Brewster Angle) and Fresnel Coefficients, Interference, Diffraction and Fermat Principle in Optics (5 Lectures)
Ray Optics: Optical Momentum, The Eikonal Equation , The Ray Equation, Point Spread Function (PSF), Optical Transfer Function (OTF) (6 Lectures)
Design of Two-Dimensional Concentrators: Thin lens formula, Concentrators for Sources at a Finite Distance and Tubular Receivers. (6 Lectures)
Radiation and Detectors: Black-body Radiation, Radiometry, Photometry, and Radiation Heat Transfer, Radiation detectors: Spectrometer, Power meter and Flux meter (7 Lectures)
Sun-earth geometry: Derivation of incidence angles and tracking algorithms for the principal solar tracking modes and Derivation of projected incidence angles for stationary concentrators (3 Lectures)
Design of Nonimaging Optics: Imaging Properties of the Design Method Nonimaging Applications, Conservation of Étendue, The Winston–Welford Design Method, Tailored Edge Ray Concentrators Designed for Maximum, Concentration , Tailored Edge, Ray Concentrators Designed for Lower Concentration , Fresnel Primaries and Applications of Nonimaging Concentrators. (7 Lectures)
Applications to Solar Energy Concentration high concentration: Design of Solar thermal fields, Design of Heliostat field, Parabolic Trough(PT), Linear Fresnel Reflector (LFR) and Design of Compound Parabolic Collectors (CPC), Ray Tracing Technique for Field Design (6 Lectures)

Textbook

1. Welford W. T and Winston, R., High Collection Non-Imaging Optics, Academic Press 1989.
2. Hecht, E., Optics, 4th Ed., Pearson Education 2002.

Reference book

1. Born, M and Wolf, E., Principles of Optics, 7th Ed., Cambridge 1999.
2. Pedrotti, F. L., Pedrotti, L.M and Pedrotti, S. L., Introduction to Optics, 3 rd Ed., Pearson 2008.
3. Donald C. O'Shea: Elements of Modern Optical Design. John Wiley & Sons, 1985
4. Julio Chaves: Introduction to Nonimaging Optics. CRC Press, 2008.

Self-Learning Material

Preparatory Course Material

Title	**Theory of Vibrations**	Number	MEL7XX0
Department	Civil Engineering	L-T-P-Th [C]	3-0-0 [3]
Offered for	M.Tech, M.Tech.-Ph.D., Ph.D.	Type	ME
Prerequisite	Basic knowledge of mechanical vibrations		

Objectives

The Instructor will:

1. Provide the approach adopted for analyzing the structural dynamic characteristics and dynamic response of uniform and non-uniform structural elements, like beams.
2. The approach is applicable for structural dynamic analysis of any structure.

Learning Outcomes

The students are expected to have the ability to:

1. Analyze the vibration characteristics of non-uniform beams.
2. Thorough understanding of approximate methods used in the structural dynamic analysis.

Contents

Introduction: Discrete and continuous systems, SDOF, Types of Damping, Frequency response, Time response, Beating phenomenon, Response to impulse loading, Convolution integral, Numerical integration, Newmark-β method, Wilson-θ method *[Lectures 9].*

Rayleigh's Method: Approximate evaluation of natural frequency, improved Rayleigh's method *[Lectures 4].*

Stiffness and flexibility influence coefficients, finite element stiffness of beam elements, Static condensation *[Lectures 5].*

Hamilton's principle, Lagrange's equations *[Lectures 3].*

Linear conservative natural systems: Symmetric eigenvalue problem, Modal analysis, Systems with proportional damping, structural damping (complex stiffness) *[Lectures 4].*

Continuous parameter systems: Longitudinal vibration of bars, bending vibration of beams, torsion vibration *[Lectures 4].*

Differential Eigenvalue Problem: Admissible function, Comparison function, Eigen function *[Lectures 2].*

Positive definite system, self adjoint property *[Lectures 2].*

Natural modes of vibration: Mode shapes and natural frequencies *[Lectures 2].*

Approximate methods: Transfer Matrix method, Rayleigh's quotient, Rayleigh-Ritz method, Galerkin method *[Lectures 5].*

Response to excitation: Mode summation, Mode acceleration *[Lectures 2].*

Reference Books

1. Meirovitch, L., Principles and Techniques of Vibrations, 3rd Edition, Prentice Hall
2. Meirovitch, L., Analytical Methods in Vibration Analysis, Macmillan
3. Clough, R.W., and Penzien, J., Dynamics of Structures, McGraw Hill

Online course Material

https://nptel.ac.in/courses/105101006/

Title	**Offshore Geo-technology**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3–0–0 [3]
Offered for	B.Tech., M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

1. To familiarize students with the dynamics of offshore environment.
2. To introduce methods of soil and site investigation.
3. To introduce designs, types, and analyses of piled foundation, shallow foundation, and anchoring systems.
4. To introduce specific consideration of soil structure interaction of on-bottom and pipelines and risers.
5. To provide overview of geo-hazards.

Learning outcomes:

1. Detailed knowledge of geophysical, geotechnical investigation, laboratory test and use of centrifuge modelling.
2. Understanding of soil response under static and dynamic loads.
3. Knowledge of the analysis and design of shallow foundation, piled foundation, anchoring system and on-bottom pipelines and risers.
4. Understanding of marine slope failure and slides.

Contents:

Offshore environment: Marine geology, Plate tectonics, Continental self, Marine-sediments, Geotechnical characteristics, Ocean wind and waves, currents, wave theories, hydrodynamic forces (Lectures 7).

Offshore site and soil investigation: Geophysical and Geotechnical investigations; Soil classifications and tests- undrained tri-axial compression test, undrained strength; cyclic loading test, failure under cyclic loading, Centrifuge modelling [Lectures 7].

Offshore foundations: Shallow and deep foundations and their applications, Design of offshore foundations as per API and DNV codes, Pile drivability and dynamic monitoring, Capacity of driven piles, Installation piled and shallow foundations in fine grained and permeable sediments, Stability of shallow foundations, Dynamic analysis of vertical and inclined pile foundations, Impedance function of piles and frequency domain analysis [lectures 10].

Anchoring systems: Buoyant platforms and mooring systems, Mooring line analysis and forces on anchors, Types of anchors and their design; API and DNV code provisions for anchor design, Dynamic effects on anchor and uplift of anchors, Dynamically penetrating anchors [Lectures 7].

Pipeline and riser geotechnics: Geotechnical inputs to pipeline design; Design issues and code provisions; Soil structure interaction for on-bottom pipelines and risers, Secondary stabilization [Lecture 6]

Offshore geo-hazard: Types of marine geo-hazard and their identification, Submarine slope failures and slides, probabilistic hazard analysis [Lectures 5].

Text Books:

1. Mark Randolph, Susan Gourvenec (2011). Offshore Geotechnical Engineering. Spon Press (Taylor & Francis), ISBN13: 978-0-415-47744-4(hbk).
2. Paolo Boccotti (2000). Wave Mechanics for Ocean Engineering (1st Ed), Elsevier Science. ISBN: 9780080543727 (eBook)

Title	**Rock Mechanics and Rock Engineering**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

1. To understand fundamentals of stress-train behavior of isotropic and anisotropic rocks under different conditions
2. To understand limits of rock strength and analysis of rock failure conditions
3. To understand the influence anisotropy under at different scales
4. Introduction of use of rock mechanics in tunneling, mining engineering, and reservoir geomechanics

Learning outcomes

1. Knowledge of geological and engineering classification of rocks.
2. Detailed knowledge of failure criteria and influence of ambient conditions
3. Knowledge of laboratory- and field-based stress measurement techniques
4. Understanding of isotropic and anisotropic rock deformation
5. Knowledge of tunneling in isotropic and anisotropic rocks
6. Knowledge of natural and artificial support system in underground excavations
7. Knowledge of wellbore stability and hydraulic fracturing

Contents

Classification and index properties of rocks: Geological classification of rocks; Index properties of rocks; Porosity; Density; Hydraulic conductivity and permeability; Strength; Slaking and durability; Sonic velocity; Classification of rock masses *[Lectures 4].*

Rock strength and failure criteria: Modes of failure; Laboratory strength tests; Stress-strain behaviors in compression; Application of complete stress-strain curve; The Mohr-Coulomb failure criterion; Hoek-Brown failure criterion; Effect of size, water, and anisotropy on strength *[Lectures 6].*

Initial stresses in rocks and their measurement: Estimating the initial stresses; Techniques for measurement of in-situ stresses; Determination of S_{Hmin} from leak-off tests; Wellbore failure and the magnitude of S_{Hmax}; Drilling-induced tensile fracture and magnitude of S_{Hmax}; Estimation of rock strength from breakouts *[Lectures 5].*

Planes of weakness in rocks: Joint orientation; Joint roughness; Relationship between displacement and strength; Effect of water pressure *[Lectures 5].*

Deformability of rocks: Elastic and non-elastic behavior; Measurement of deformability properties by static tests; Measurement of deformability properties by dynamic measurement; Fractured rock; Influence of time on rock deformation *[Lectures 6].*

Rock mechanics for tunneling: Opening in competent rocks; Horizontally layered rocks; Rock with inclined layers; Plastic behavior around tunnels; Time dependent behavior of tunnels *[Lectures 5].*

Rock mechanics for mining engineering: Natural support: Pillar design approach, Pillar strength equation, Model of pillar failure, Pillar stabilization; Artificial support: Rock support interaction analysis, Classification, Primary reinforcement, Secondary support system; Rock blasting process, control, and optimization *[Lectures 5].*

Rock mechanics for hydrocarbon reservoir: Tectonic stress field; Pore pressure measurement; Mechanical properties and stress data from laboratory analysis and field data; Concepts of critically stressed crust; Basics of hydraulic fracturing; Wellbore stability *[Lectures 6].*

Text Books

1. Richard E. Goodman (1989) Introduction to Rock Mechanics (2nd Ed). Wiley. ISBN-10: 0471812005.
2. Mark D. Zoback. (2007) Reservoir Geomechncs. Cambridge University Press. ISBN-978-0-521-77069-9.
3. Debasis Deb, Abhiram Kumar Verma (2016) Fundamentals and Applications of Rock Mechanics. PHI Learning Pvt. Ltd. ISBN-978-81-203-5182-0.

Online course Material

1. V. B. Maji. Rock Mechanics – Web course. https://nptel.ac.in/courses/105106055/

Title	**Stability of slopes**	Number	**CIE7XX0**
Department	Civil and Infrastructure Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives:

1. To understand the influence of geology, rockmass properties, and hydrology on the stability of slopes.
2. To present basic types of analytical and numerical methods of slope stability analysis.
3. To introduce methods and instruments of slope monitoring and stabilization.
4. To enable planning and design of preventive measures, and redesign of failed slopes.

Learning outcomes:

1. Knowledge of data collection from field-work, aerial photography, geophysics, and geological map.
2. Knowledge of *kinematic analysis* of slopes using geological data and stereographic projection.
3. Knowledge of *kinetic analysis* of slopes and influence of critical geometrical and hydrological parameters.
4. Knowledge of available methods of slope stabilization by reinforcements and rock removal.
5. Knowledge of instrumental facilities to monitor slope movement and subsequent classification.

Course content:

Structural geology for slope analysis: Geological structures; Orientation of discontinuities; Introduction to geological maps; Identification of modes of slope instability [*Lecture:* 4]

Site investigation and geological data collection: Site reconnaissance; Geological mapping; Spacing, persistence, and roughness of discontinuities; Probabilistic analysis of structural geology; Drilling and logging. [*Lecture:* 4]

Rock strength properties: Scale effect and rock strength; Shear strength of discontinuities; Laboratory testing of discontinuities; Hoek-Brown strength criterion for fractured rock mass; Rock durability and compressive strength [*Lecture:* 4]

Plane failure: Conditions of plane failure; Influence of groundwater on stability; Critical tension crack: identification and influence; Influence of critical slide plane inclination; Analysis of failure on a rough plane; Reinforcement of a slope; Seismic analysis of slopes [*Lecture:* 5]

Wedge failure: Conditions of wedge failure; Wedge analysis including cohesion, friction, and water pressure; Wedge stability chart for friction; Compression wedge analysis [*Lecture:* 5]

Circular failure: Conditions of circular failure; Methods of analysis; Derivation and use of circular failure charts; Location of critical slide surface and tension crack; Bishop's and Janbu's method of failure analysis. [*Lecture:* 6]

Toppling failure: Types of toppling failure; Kinematic analysis of block toppling failure; Limit equilibrium analysis of toppling [*Lecture:* 5]

Stabilization of slopes: Stablization programs: Rock slope Inventory, Hazard rating analysis; Stabilization by rock reinforcements; Stabilization by rock removal; Measures against rock fall [*Lecture:* 5]

Movement monitoring: Types of slope movement: Initial response, Regressive and progressive movements, Long-term creep; Surface monitoring methods; Data interpretation [*Lecture:* 4]

Suggested Textbooks:

1. Duncan C Wyllie, Christopher W. Mah (2005) *Rock Slope Engineering*, 4th Ed, Spon Press (Taylor & Francis Group), ISBN 0–415–28001–X (pbk).
2. Robin Chowdhury, Phil Flentje, Gautam Bhattacharya (2010) *Geotechnical Slope Analysis*. CRC Press (Taylor & Francis Group), ISBN: 978-0-415-46974-6 (Hbk).

Title	**Wind and Earthquake Engineering**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B.Tech, M.Tech, M.Tech.-Ph.D., Ph.D.	Type	ME
Prerequisite			

Objectives

1. To acquaint the students with the fundamental knowledge of wind and earthquake engineering to pursue higher level courses/research on the wind and earthquake engineering
2. To provide the requisite information on the above two subjects to the students to work in design offices.
3. To impart the back ground materials for pursuing some courses offered in the fields of specialization (cores).

Learning Outcomes

1. Enable the students to apply the knowledge gained in solving practical problems.
2. Enable the students to analyze the collected wind and earthquake data for providing systematic inputs to be used in practice.
3. To make the students capable of using the knowledge gained for wind and earthquake hazard mapping.

Contents

Atmospheric wind circulation, its mechanism, types of wind, wind measurements, wind characteristic, different ways to express wind energy, inputs required for the analysis of structures, aero dynamic and aero-elastic effects, aero elastic instabilities-flutter, galloping, torsional divergence, dynamics in the wind codes *[lectures 20].*

Inside of the earth and tectonic plate movement, major tectonic plates, fault lines, tectonic theory of the occurrence of the earthquake, earthquake hazards, earthquake input for structural analysis, concept of the response spectrum, deterministic seismic analysis of structures, salient features of earthquake code *[lectures 20].*

Text Books

1. Ghosh S K and Fanella D A, Seismic and Wind Design of Concrete Buildings, International Code Council-New York (2004).
2. Taranath B S, Wind and Earthquake Resistant Buildings: Structural Analysis and Design, Marcel Dekker, New York (2004).

Others:

1. Datta T K, Seismic Analysis of Structures, John Wiley & Sons, New York (2010).
2. IS 1893-Part-I, Criteria for Earthquake Resistant Design of Structures: General Provisions and Buildings, Bureau of Indian Standards, New Delhi (2016).

Title	**Process safety and Hazards**	Number	CH7XX0
Department	Chemical Engineering	L-T-P [C]	3-0-0
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite	None		

Objective

The Instructor will:

1. Provide comprehensive knowledge of safety regulations and their practices, chemical plant hazards and their control, risk assessment and management principles and techniques and accident analysis.

Learning Outcomes

The students are expected to have the ability to:

1. Identify and distinguish typical sources of risk and hazard in a process plant, undertake hazard and operability (HAZOP) study.

Contents

Safety in Industries: Concepts and definition, storage of dangerous materials, Plant layout Safety systems, Occupational Safety and Health Administration (OSHA) incidence rate, Fatal accident rate (FAR), Measures to take during accidents, Toxicology (10 lectures)

Industrial Hygiene: Government regulations, Industrial hygiene and safety aspects related to toxicity, noise, pressure, temperature, vibrations, radiation etc, Evaluation Methods, Evaluating workers Exposures to dusts, noise. (8 lectures)

Selection Mechanism: Technology and process selection, Scale of disaster, Fire triangle Ignition, Confined and unconfined explosion, Vapors cloud explosions, Boiling liquid expanding vapor explosion (BLEVE), Dust explosion, shock wave, Flammability characteristics of liquids and vapors, Minimum oxygen concentration (MOC) and Inerting. (8 lectures)

Handling Chemicals and toxicity: Control of toxic chemicals, Storage and handling of flammable and toxic chemical, Runway reactions, Relief system risk and hazards management, Design to prevent Fires and Explosions, Miscellaneous Design for preventing Fires and Explosion. (8 lectures)

Hazards Identification: checklists, Surveys, Hazard and Operability Studies (HAZOP), Safety reviews. Risk Assessment, Hazard models and risk data. Tackling disasters, plan for emergency. Risk management routines, Emergency shutdown systems, Role of computers in safety, Prevention of hazard human element, Technology and process selection. (8 lectures)

Text Books

1. Daniel A. Crowl and Joseph F. Louvar, 1990, *Chemical Process Safety: Fundamentals with applications,* Prentice Hall, Inc.
2. Lee, F.P., 1983, *Loss prevention in the process Industries*, Volume 1 and 2 Butterworth.
3. Hoboken, N. J., 2000, *Guidelines for Chemical Process Quantitative Risk Analysis*, Wiley-Interscience.

Reference Books

1. R. W. King and J. Magid, *Industrial Hazards and Safety Handbook*, Butterworth, 1982.
2. G. L. Wells, *Safety in Process Plant Design*, John Wiley and Sons Inc., 1980.
3. Fawcett, H.H. and W.S. *Wood, Safety and Accident Prevention in Chemical Operations*, 2nd Edition, Wiley-Interscience, New York, 1982.

Title	**Advanced Energy Systems**	Number	CH7XX0
Department	Chemical Engineering	L-T-P [C]	3-0-0
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite	None		

Objectives

The Instructor will:

1. Provide comprehensive knowledge about conventional and several advanced energy production technology.

Learning Outcomes

The students are expected to have:

1. Good knowledge about different energy production systems.

Contents

- Introduction: Latest topics on energy, Energy fuels for power generation, conventional power generation technology. [6 lectures]
- Emerging clean energy systems: Integrated Gasification combined cycle (IGCC), Pressurized fluidised bed combustion (PFBC, AFBC). [6 lectures]
- Heavy fuel based power generation. [3 lectures]
- Advanced combustion systems. [4 lectures]
- Understanding Natural gas cycles. [4 lectures]
- Energy conservation in power plants. [4 lectures]
- Geological CO_2 sequestering. [3 lectures]
- Natural gas cycles. [4 lectures]
- Materials-based Solutions to advanced Energy systems: role of advanced material, general devices and process approaches, key materials for advanced energy systems, structural material. [8 lectures]

Textbook

1. Khartekenko, N., Khartekenko, V., 2010, *Advanced Energy Systems*, Taylor & Francies, Second Edition.

Reference Books

1. Colin Tong, 2019, *Introduction to Materials for Advanced Energy Systems*, Springer.
2. I. Dincer and C. Zamfirescu, 2011, *Sustainable Energy Systems and Applications*, Springer.

Title	**Microfluidic Fuel Cells**	Number	CH7XX0
Department	Chemical Engineering	L-T-P [C]	1-0-0
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite	None		

Objectives

The Instructor will:

1. Provide in-depth knowledge about the fundamentals of microfluidics devices as fuel cells.

Learning Outcomes

The students are expected to have:

1. Detailed understanding of Microfluidic fuel cell.

Contents

Microfluidic cells: Introduction, concept, classification, and challenges, categories, Thermodynamic and kinetic of microfluidic fuel cells, electrochemical principles and transport. [7 lectures]

Designs: Microfabrication techniques, 2D, 3D electrodes in microfluidic fuel cells, performance evolution. [3 lectures]

Microfluidics devises: co-laminar streaming, fuel cells with biocatalysts, gas diffusion electrodes, microfluidics redox batteries, cell arrays and Stacks. [4 lectures]

Textbook

1. Erik Kjeang, 2014, *Microfluidic Fuel Cells and Batteries,* Springer.

Reference Books

Research Articles from different journals.

Title	**Air Pollution and Control**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

1. To gain understanding of basic concept of air pollution, its sources and effects on human health and climate.
2. To introduce concepts of aerosol dynamics used in ambient sampling, designing measurement technologies and air quality modeling.
3. To develop understanding of the concepts and strategies for control of particulate matter and gas-phase pollutants.
4. To provide knowledge of current state of the atmosphere and introduce fundamentals of pollutant transport and dispersion, and modeling.

Learning Outcomes

The students are expected to achieve proficiency in:

1. Developing a broad overview of local-to-global scale pollution issues and their specific impact on human health and environment.
2. Solving problems related to dispersion and air quality modeling
3. Understanding the source level air pollution issues and identify the right control techniques/measures to meet the desired standard levels.
4. Computing the performance of various unit operations and processes used in the air treatment.

Contents

Introduction to air pollution: Historical events [Meuse River Valley 1930, Donora Pennsylvania 1948, London Smog 1952, Bhopal Gas Tragedy 1984, and related Concepts], Scales of air pollution, Sources and classification [CAPs and HAPs] of air pollutants, Air quality and standards, Emission Inventory, The Earth's atmosphere: Structure and Composition, and energy balance *[Lectures 6].*

Aerosol Physics: Fundamentals: Source, Size and Size distribution, Gas and Particle motion: Brownian motion, Diffusion, Inertia, External force field, Aerosol dynamics: coagulation, condensation, nucleation, particle measurements and Sampling techniques *[Lectures 10].*

Emissions Control devices: Removal of particles from gas stream: Settling chambers, Cyclone separators, electrostatic precipitation, filtration, wet collectors, collection efficiency; Removal of gases from effluents stream: Absorption by liquids, adsorption on solids, Examples: SO_2 and NO_x removal *[Lectures 8].*

Meteorology: Atmospheric meteorology, Wind velocity profile and Wind rose, turbulence, inversion layer and mixing depth and Atmospheric Stability *[Lectures 6].*

Dispersion of air pollutants and Air Quality modeling: Dispersion models, Gaussian dispersion model: assumptions, dispersion parameters and effective stack height, plume rise and plume behavior, Receptor models *[Lectures 6].*

Effects of Air Pollution on Climate and Human health: Tropospheric Ozone chemistry, Photochemical Smog, Acid Rain, Visibility, Indoor Air Pollution, Dose Response and Health effects *[Lectures 6].*

Text Books

1. Atmospheric Chemistry and Physics- From Air Pollution to Climate Change, J. H. Seinfeld and S. N. Pandis, John Wiley and Sons, Inc., Third Edition, 2013.
2. Aerosol Technology-Properties, Behavior and Measurement of Airborne Particles, W. C. Hinds, John Wiley and Sons, Inc., Second Edition, 1998.
3. Introduction to Air Pollution Science, R. F. Phalen and R. N. Phalen, Jones & Bartlett Learning, LLC, Burlington, MA, 2013.
4. Aerosol Measurement: Principles, Techniques and Applications, edited by K. Willeke, P. Baron, Van Nostrand Reinhold, 1993.

Online course Material

https://nptel.ac.in/courses/105104099/

Title	**Solid Waste Management**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

1. To enable understanding of life cycle of waste generation to disposal or sustainable re-use
2. To inculcate basic steps of solid waste management
3. To enable selection of best possible location specific solid waste management options

Learning outcomes

1. Conceptually design waste conversion/treatment processes
2. Able to assess the environmental and the economic impact of specific types of solid waste generated and its reuse
3. Enabled to develop innovative and smarter solutions of solid waste management at rural and urban environs

Contents

Solid waste type and Classification, Quantity, Types of Waste Stream and its life cycle Waste Generation- Industrial, Municipal and Agriculture, 4R strategies and Social costs *[Lectures 4].*

Collection and Transport: Collection Components, their Reliability, Scavengers, Automated Waste Collection, Design for Larger Transfer Operations, Automated Control, Inventory and Monitoring *[Lectures 5].*

Separation and Processing Techniques: Separation and Screening Equipment, Automated Sorters, Solid Waste Processing, Recovering, E-Waste, Size Reduction, Incineration and pyrolysis *[Lectures 6].*

Options for Disposal, Selection Criteria, Sanitary Landfill, Properties of Gases from Landfills, Hazards and Air Pollution, , Leachate Control and Management *[Lectures 5].*

Composting: Benefits of Composting, Biological Processes Involved in Composting Chemical Processes, Physical Processes, Thermodynamics of Compositing, Gasifying, Refuse-derived fuel *[Lectures 8].*

Microbiology: Biological decomposition, Microbiology of composting, Municipal solid waste Pathogens, enteric microbes in landfills and leachate, Pure culture, Mixed culture screening tests *[Lectures 6].*

Reuse and Recycle: Reuse, Increased Product Lifetime, Material Life Cycle Studies, Recycling, New Materials-Self Destructing/Organic, MSW Handling Rules, Public Aptitude, and Legislation *[Lectures 6].*

Text Books

1. George Tchobanoglous, Hilary Theisen, Samuel Vigil, Integrated Solid Waste Management: Engineering Principles and Management Issues, McGraw-Hill Series in Water Resources and Environmental Engineering, ISBN-10: 0070632375, ISBN-13: 978-0070632370
2. Kanti L. Shah, Basics of Solid and Hazardous Waste Management Technology, Prentice Hall, ISBN-10: 0139603786, ISBN-13: 978-0139603785
3. Ni-Bin Chang, Ana Pires, Sustainable Solid Waste Management: A Systems Engineering Approach, Wiley-IEEE Press; 1 edition (March 16, 2015), ISBN-10: 1118456912, ISBN-13: 978-1118456910

References Books

1. Edward A. McBean, Frank A. Rovers, Grahame J. Farquhar, Solid Waste Landfill Engineering and Design, Prentice Hall; ISBN-10: 0130791873, ISBN-13: 978-0130791870
2. Christian Ludwig, Stefanie Hellweg, Municipal Solid Waste Management , Springer 2003, London, ISBN 978-3-642-62898-6.
3. Roger Tim Haug, Composting Engineering: Principles and Practice, Ann Arbor Science, MI, ISBN-0-250-40347-1

Online Resources

T V Ramachandra Municipal Soild Waste Management, IISc Bangalore https://nptel.ac.in/courses/120108005/

Solid and Hazardous Waste Management,IITM, https://nptel.ac.in/courses/105106056/

Title	**Remote Sensing and GIS for Environment Management**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P-D [C]	3-0-2 [4]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

1. To introduce you to the basic principles of observing the Earth's geography and concepts of remote sensing
2. To explore its applications specific to the environment and natural resource management.

Learning Outcomes

1. Identify and obtain appropriate satellite imagery for a land surface based resource management, service infrastructure and real world application
2. Demonstrate proficiency in the common (open source: QGIS/ Google Earth Engine, ERDAS Imagine, ENVI) and advanced (ARCGIS, Ecognition, QT modeler) remote sensing software packages
3. Conduct basic to intermediate remote sensing analyses (atmospheric and geographic corrections, spectral transformations and enhancements, classification, modelling and change detection)

Contents

In-situ field observation; data collection; Photographic and Remote Sensing data collection, Spectral spatial and temporal aspects, analog and digital processing introduction and sensing economics [Lectures 4]
Electromagnetics: conduction, convection and radiation; radiation and its models, [Lectures 6].
Aerial Photography- Camera, photography filtration-films, planning field missions [Lectures 4]
Photogrammetry : Aerial photography, stereoscopy and its principles, image digitization, aero-triangulation, extracting spot elevation, contours and DEMs using soft copy photography, Ortho-imagery and Area Measurement [Lectures 6]
Multispectral Imagery: discrete detectors and scanning mirrors, Landsat: multi-spectral scanner, thematic mapper, Enhanced Thematic Mapper plus (Landsat 7), , GOES-,AVHRR Radiometer, Aircraft Multispectral Scanners [Lectures 8]
Thermal Infrared Remote Sensing: Thermal infrared radiation properties, laws (Stefan-bolzmann, Wein's, Kirchoff's)[Lectures 2]
Microwave Remote Sensing: RADAR, SAR remote sensing, Radar interferometry, LIDAR Passive Microwave Remote Sensing- Radiometers. [Lectures 3]
Visual image Interpretation: The aerial/regional perspective, 3D depth perspective [Lectures 5]
Digital Image Interpretation and Analysis: Preprocessing of Images, Image Enhancement, Contrast and basics of spatial feature manipulation. [Lectures 5]

Note: Labs will have hands-on use of the latest Google Earth Engine (GEE), QGIS, ERDAS IMAGINE, and ArcGIS software. It will also look into application of remote sensing imagery in soil and water analysis, overlaying data retrieved from different subject areas of engineering, lidar imagery and other GIS data.
List of Experiments on Surveying Based Laboratory:

1. To assess the distance between two inaccessible points in a specific area using a compass as well as plotting the area using closed traverse (prismatic compass).
2. To solve two-point, and three-point problems using plane table survey.
3. To understand differential levelling, vertical/cross sectional plotting and contouring using automatic levellers.
4. To survey a nearby plot bounded by any polygon using a theodolite and also to perform trigonometric levelling to find the characteristics of a nearby building.

5. To plot a known area using GPS measurements and assess the accuracy of the measurement
6. Using total station to combine electromagnetic distance measurement and electronic theodolite to understand the ramp in the institute and their geometrical features with high accuracy.

List of Experiments in GIS Laboratory:

1. To create a map of any area using vector data models using available ground data.
2. To perform geo-referencing when coordinate values are not given.
3. To study terrain data such as slope, aspect, and hill shade. How to use these for representation of different social/service as well as frequency of different aspects concerning demographic data.
4. To create a stream network or perform stream order generation or delineate watersheds created by a water body using DEM files.
5. To perform site suitability analysis for a service building (example hospital) near to a agricultural and inhabited land near an arid village at the edge of an oasis near Jodhpur.
6. Basic experiments on image sharpening and image resolution using ARC GIS/ ERDAS IMAGINE processing tool.

Text Books

1. John R. Jenkins, Remote Sensing of the Environment: An Earth Prespective
2. Lillesand, T.M., Kiefer, R.W. & Chipman, J.W. 2008. Remote Sensing and Image Interpretation, 6th Edition, John Wiley and Sons.

Reference Books

1. Floyd F. Sabins, Remote Sensing: Principles and Applications (3rd edition), Publisher : Waveland Press, Edition : 3rd, Print ISBN : 9781577665076, 1577665074, eText ISBN : 9781478618171.0
2. Gopi, S., Sathikumar, R., and Madhu, N., Advanced Surveying: Total Station, GIS and Remote Sensing, Pearson, 2007, Dorling Kindersley (India).ISBN 978-81-3017-0067-9.
3. Barrett, E.C. & Curtis, L.F. 2007. Introduction to Environmental Remote Sensing, Routledge Publisher.

Title	**Climate Change and Impact**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

1. To gain understanding of scientific foundation and key concepts of atmospheric science, climate data collection and impacts of GHGs.
2. To develop scientific basis of our understanding of climate system and climate change, and associated uncertainties.
3. To provide knowledge on the impacts of climate change on natural world in the current scenario and coming centuries.

Learning Outcomes

The students are expected to achieve proficiency in:

1. Understanding present climate system in the context of Earth's history, reconstructing climate evolution and relevant scientific uncertainties.
2. Developing basics of climate change science with key concepts of climate, weather, energy balance and GHG effect.
3. Understanding energy flows within Earth-Atmosphere system.
4. Explaining the large-scale atmospheric circulations with underlying fluid mechanics principles and role of oceans in climate variability.
5. Developing scientific understanding of climate forcing, sensitivity and feedback systems.
6. Understanding the basics of energy balance models and fundamental equations of Global Circulation Models for simulating and estimating past and future climate systems.
7. Understanding the potential impacts of climate change and future climate change projections.

Contents

Introduction to Climate and Climate Change: Evolution of Earth's climate: history, Instrumental and paleoclimatic record, Climate Change: Definition, Components of climate system, Global energy balance, GHGs and Greenhouse effect *[Lectures 5].*

Radiative Energy Transfer: Radiation laws, Radiation Processes, Absorption and Scattering by Gases and Particles, Solar Zenith Angle and Aerosol Optical Depth, Radiative Transfer Equation *[Lectures 5].*

Convection and Atmospheric Circulation: Dry and moist convection, Atmospheric convection and types, Scales of motion, Energy and Momentum Transport *[Lectures 7].*

Oceanic Circulation: Physical characteristics of ocean, Energy transport mechanism: Wind driven circulation theories, and thermohaline circulation *[Lectures 4].*

Climate sensitivity and feedback mechanisms: Equilibrium Climate sensitivity, Transient Climate Response, Feedback: Concept, Fast and slow feedback mechanisms/Processes, Climate sensitivity estimates *[Lectures 5].*

Climate Variability: Buffering atmospheric temperature changes, Atmosphere-ocean coupling: El Nino and Southern Oscillations *[Lectures 3].*

Climate modeling: One layer energy balance model, 1D EBM, Global Circulation Model, difference between Weather forecast models and GCMs *[Lectures 8].*

Climate Change Impacts and future projections: Sea level rise, Ecosystem and biodiversity, Shifting water resources, Agricultural impacts, severe weather impacts and Human health; Impact assessment of GHGs on global temperature, rainfall and drought patterns, atmospheric circulation. *[Lectures 5].*

Text Books

1. Atmospheric Chemistry and Physics- From Air Pollution to Climate Change, J. H. Seinfeld and S. N. Pandis, John Wiley and Sons, Inc., Third Edition, 2013.
2. Fundamentals of Atmospheric modeling, Marck Jacobson, Cambridge University Press, Second Edition, 2005.
3. Air Pollution and Global Warming: History, Science and Solutions, Mark Jacobson, Cambridge University Press, Second Edition, 2012.
4. Atmosphere, Ocean, and Climate Dynamics, John Marshall ad R. Alan Plumb, Elsevier Academic Press, 2008.

Online course Material

https://courses.edx.org/courses/course-v1:SDGAcademyX+CCSI001+3T2019/course/
https://nptel.ac.in/courses/119106008/

Title	**Green building and Sustainable Materials**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives: Objectives of this course are:

1. To introduce elements of green building construction and use of sustainable materials.
2. To familiarize students with different national and international green performance ratings.
3. To teach students methods of carbon footprint reduction and increase in energy efficiency of a building.

Learning Outcomes: A graduate of this course will have a thorough idea of:

1. National and international green building assessment system and requirements.
2. Green building design elements and use of sustainable materials.
3. Techniques to reduce the carbon footprint of a green building.
4. Management of hydrological cycle, treatment of waste, and maintainance of indoor air quality.
5. Green building economics, business plan, and management.

Contents

Introduction to green building: Rationale for high-performance green building; green building progress and obstacles; trends in high-performance green building; the green building movement. *(Lectures: 3)*

Ecological design: design vs ecological design; contemporary ecological design; future ecological design; thermodynamics: limits on recycling and the dissipation of materials *(Lectures: 3)*

Green building assessment: US green building assessment system; international green building assessment system; performance assessment. *(Lectures: 3)*

LEED building rating system: history of LEED; structure of the LEED suite of building assessment systems; LEED credentials; the LEED process; LEED categories. *(Lectures: 3)*

The Green Globes building assessment system: the Green Globes process; Green Globes verification and certification; structure of the ANSI/GBI 01-2010 standard; the Green Globes professional and Green Globes Assessor. *(Lectures: 3)*

Green building design process: conventional vs green building delivery systems; executing the green building project; the integrated design progress; role of Charrette in the design process; green building documentation requirements. *(Lectures: 3)*

Sustainable site and landscape: land and landscape approaches for green buildings; sustainable landscapes; enhancing ecosystems; stormwater management; low-impact development; heat island mitigation; light trespass and pollution reduction; sustainable site initiative. *(Lectures: 3)*

Energy and carbon footprint reduction: building energy issues; high-performance building energy design strategy; passive design strategy; building envelope; internal load reduction; active mechanical system; water-heating system; electrical power systems; innovative energy optimization strategies; renewable energy systems; fuel cells; smart buildings and energy management systems; ozone-depleting chemicals in HVAC&R systems; reducing carbon footprint of the built environment. *(Lectures: 4)*

Built environment hydrological cycle: global water resource depletion; hydrological cycle terminology; high-performance building hydrological cycle strategy; designing the high-performance building hydrological cycle; water budget rules of thumb; sustainable stormwater management; landscaping water efficiency. *(Lectures: 4)*

Closing materials loops: distinguishing between green building products and green building materials; LCA of building materials and products; environmental product declaration; materials and product certification systems; emerging construction materials and products; design for deconstruction. *(Lectures: 4)*

Indoor environmental quality: IEQ issues; design; main components of integrated IEQ design; emission from building materials; economic benefits of good indoor environmental quality. *(Lectures: 3)*

Construction operations and commissioning: site protection planning; indoor air quality during construction; construction materials management; construction and demolition waste management. *(Lectures: 3)*
Green building economics: the business case for high-performance green building; the economics of green building; quantifying green building benefits. *(Lectures: 3)*

Text Books

1. Charles *J.* Kibert (2013). *Sustainable Construction*. John Wiley & Sons. ISBN 978-1-118-33013-5
2. J. Newman and Ban Sang Choo (2003) *Advanced Concrete Technology-Constituent Materials (1st Ed)*, Elsevier. ISBN 9780080489988.
3. S. Kubba (2010) *LEED Practices, Certification, and Accreditation Hand book (1st Ed)* Elsevier. ISBN 9780080958590.

Additional materials:

1. Ministry of Power, Energy Conservation Building Code 2018, Revised Version, Bureau of Energy Efficiency, 2018,
2. Indian Building Congress, Practical Handbook on Energy Conservation in Buildings, 1 st ed. Nabhi Publication, 2008.
3. TERI-Griha's Green Design practices (www.teriin.org/bcsd/griha/griha.htm)
4. Sartori, I., and Hestnes, A. G. "Energy use in the life cycle of conventional and low-energy buildings: A review article", Energy and Buildings., 20, 249-257.2007
5. Green Building Basics, California Integrated Waste Management Board (www.ciwmb.ca.gov/GREENBUILDING/Basics.htm#What)
6. Catarina Thormark. "A low energy building in a life cycle—its embodied energy, energy need for operation and recycling potential", Building and Environment., 37, 429-435.2001.

Online course Material
"*NOC:Energy Efficiency, Acoustics and daylighting in Building*" by Prof. B. Bhattacharjee on NTPEL. https://nptel.ac.in/courses/105/102/105102175/

Title	**Traffic Flow Theory**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives: The course will introduce students to:

1. Fundamentals of traffic flow
2. Analysis of traffic flow
3. Traffic flow modelling approaches

Learning Outcomes: At the end of the course, the student should be able to:

1. Differentiate and analyse traffic flow parameters at micro-macro levels
2. Analyse congestion propagation in traffic flow
3. Understand the state-of-the-art traffic flow modelling
4. Perform traffic flow modelling and analysis using field data

Contents

Introduction to Traffic Flow theory [No. of lectures – 6]: Macroscopic parameters – Space mean speed, Density, and Flow; Microscopic parameters – Time mean speed, Time headway, Space headway, occupancy; Relation between microscopic and macroscopic parameters; Fundamental traffic flow equation

Macroscopic models of Traffic flow [No. of lectures -10]: Empirical and theoretical models, Continuity equations, Solution to linear and non-linear partial differential equations, Two fluid model

Microscopic Models of Traffic flow [No. of lectures-10]: Car following models, Response Stimulus model, Psychophysical model, stability, Advanced car following models

Propagation of disturbance [No. of lectures – 6]: Non-linear traffic waves and acceleration of stream, Shockwave theory and applications

Analysis of traffic flow [No. of lectures-10]: Accumulation diagrams, Incidents, Probabilistic analysis of traffic flow, Gap acceptance theory, Queuing theory and applications.

Text Books

1. Ni, D. Traffic flow Theory. BH, Elsevier, MA, USA.
2. Adolf D. May. Traffic flow fundamentals. Copyright: Prentice Hall, Inc
3. Nicholas J Garber and Lester A Hoel. Traffic and Highway Engineering. Copyright: Brooks/Cole-Wadsworth Group\Thomson learning
4. Fred. L. Mannering, Walter P. Kilareski, and Scott S. Washburn. Principles of Highway Engineering and Traffic Analysis. Copyright: John Wiley and Sons, Inc.
5. C. Jotin Khisty and B. Kent Lall. Transportation Engineering. Copyright; Pearson Education, Inc.

Additional references:

1. Kadiyali. L. R. (2016).Traffic Engineering and Transport Planning. Khanna Publishers
2. Chakroborty, P., Animesh Das (2017). Principles of Transportation Engineering. Second Edition, PHI Learning Private Limited
3. Saxena S. C. (2014). Textbook of Highway and Traffic engineering. CBS
4. Samuel Morgan (2016) Traffic Engineering and Transport Planning. Williford Pr.
5. Jaume Barceló. (2010) Fundamentals of Traffic Simulation. Springer.
6. Roess, R.P., Prassas E. S., McShane W. R. (2010) Traffic Engineering. Pearson

Title	**Traffic System and Management**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives: The course will introduce students to:

1. Traffic system characteristics
2. Traffic control system
3. Traffic studies and basic analysis
4. Intelligent Transportation System (ITS)

Learning Outcomes: At the end of the course, the student should be able to:

1. Understand in-depth the data collection methods and analysis of various traffic flow parameters
2. Design traffic control systems
3. Have adequate exposure to various traffic management measures
4. Understand the fundamentals and applications of ITS

Contents

Traffic Systems Characteristics [No. of lectures- 6]: User, Vehicle and Traffic characteristics

Traffic Studies and Basic Analysis [No. of lectures-10]: Speed, journey time, and delay studies, Volume and occupancy studies, Origin-destination Studies, Parking studies, Accident Studies. Capacity and Level of Service (LOS) analysis

Traffic Control System [No. of lectures -12]: Road markings – Principles of road markings, carriageway markings and object markings; Traffic signs – Importance of traffic signs, Types of traffic signs; Intersection control, Signal design- Illustration of signals, fixed time signals, signal design using IRC method and HCM method

Traffic Management [No.of lectures-8]: Strategies and techniques, Traffic calming, congestion studies and road pricing.

Intelligent Transportation system [No. of lectures – 6]: User services and architecture, standards and evaluation, Public Transport and bus priority, Artificial intelligence in advanced traffic and ITS

Text Books

1. Nicholas J Garber and Lester A Hoel. Traffic and Highway Engineering. Copyright: Brooks/Cole-Wadsworth Group\Thomson learning
2. Fred. L. Mannering, Walter P. Kilareski, and Scott S. Washburn. Principles of Highway Engineering and Traffic Analysis. Copyright: John Wiley and Sons, Inc.
3. C. Jotin Khisty and B. Kent Lall. Transportation Engineering. Copyright; Pearson Education, Inc.
4. https://coeut.iitm.ac.in/ITS_synthesis.pdf
5. Sussman, J. Perspectives on Intelligent Transportation Systems (ITS). New York, NY: Springer, 2005

Additional references:

1. Kadiyali. L. R. (2016).Traffic Engineering and Transport Planning. Khanna Publishers
2. Chakroborty, P., Animesh Das (2017). Principles of Transportation Engineering. Second Edition, PHI Learning Private Limited
3. Roess, R.P., Prassas E. S., McShane W. R. (2010) Traffic Engineering. Pearson

Title	**Soil Stabilization and Ground Improvement**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives: This course will introduce:

1. Scopes of soil stabilization and ground improvement in various geotechnical projects.
2. Different conventional and advanced method of ground improvement and their applications.
3. Role of ground water and soil types on the stability of soil covers and grounds.

Learning Outcomes: A graduate of this course will have a thorough knowledge of:

1. Behaviour of soils under different environmental conditions and their compaction behavior.
2. Role and planning of drainage to enhance stability of grounds.
3. Different mechanical, chemical, and cmenting stabilization techniques regularly used in stabilization programs.
4. Use of geosynthetics and influence of their properties on ground improvement.
5. Elements of grouting and injection to increase soil stabilization.

Contents

- *Formation and development of ground:* Formation of rock, soil, and soil profile; Soil distribution in India; alteration of ground after formation; reclaimed soil; ground improvement potential *(Lectures: 3)*
- *Compaction:* compaction mechanics; field procedure; surface compaction; selection of field compaction procedures; compaction quality control. *(Lectures: 3)*
- *Drainage methods:* seepage; filter requirements; groundwater; methods of dewatering systems; design steps for dewatering systems; drains. *(Lectures: 5)*
- *Precompression and vertical drains:* compressibility of soils and consolidation; pre-loading and surface fills; vertical drains; granular piles; dynamic consolidation; consolidation by electro-osmosis. *(Lectures: 5)*
- *Vibration methods:* vibro-compression; vibro-displacement compression. *(Lectures: 5)*
- *Grouting and injection:* aspects of grouting; grouting procedure; applications. *(Lectures: 5)*
- *Mechanical, cementing, and chemical stabilization:* requirements of soil stabilization; mechanical stabilization; Portland cement stabilization; bituminous stabilization; chemical stabilization; construction methods. *(Lectures: 4)*
- *Geosynthetics:* geosynthetic types; properties of geosynthetics; applications of geosynthetics *(Lectures: 5)*
- *Contemporary soil improvement techniques:* soil reinforcement; soil nailing; thermal methods; backfill materials for retaining walls; improving rock stability and quality. *(Lectures: 5)*
- *Case histories of ground improvement:* traditional ground improvements adopted in weak soils in India and abroad; codal suggestions. *(Lectures: 2)*

Text Books

1. Hausmann, M.R. (1990). Engineering Principles of Ground Modification, McGraw-Hill International Editions. ISBN: 9780070272798.
2. Dr. Purushottama Raj (2005). Ground Improvement Techniques. Laxmi Publications. ISBN: 9788170088370.
3. Bowles J.E. (1997). Foundation Analysis and Design (5th Ed). Tata McGraw-Hill International Edition. ISBN 0-07-912247-7.

Online course Material

1. Ground Improvement Techniques by Dr. G.L. Sivakumar Babu on NTPEL. https://nptel.ac.in/courses/105/108/105108075/

Title	**Foundation Engineering**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives: this course aims at :

1. Introducing basic principles and methods of foundation design and constructions.
2. Introducing different ambient conditions that influence the stability of a foundation in short and long-term.
3. Introducing methods of stable foundations in earthquake prone areas.

Learning Outcomes: Students will have the ability to

1. Analyze the properties of soil and rock to design appropriate foundations.
2. Analytical knowledge of short-term and long-term consolidation behavior of foundations.
3. Device methods to protect new and historical foundations against decay and collapse.
4. Design earthquake resistant foundation in different soils and rocks.
5. Apply international and national codes to study soil behavior and design foundations.

Contents

- *Shallow and deep foundation:* selection of foundation type; shallow foundations; deep foundation. *(Lectures: 3)*
- *Bearing capacity of foundations:* Bearing capacity of shallow foundations; bearing capacity of deep foundations; lateral load capacity of deep foundations *(Lectures: 3)*
- *Settlement of foundations:* Collapsible soil; settlement of cohecionless soil; other common causes of settlement; foundation on rock; allowable settlement. *(Lectures: 3)*
- *Consolidation:* laboratory consolidation test; immediate settlement; primary consolidation; rate of primary consolidation; secondary compression; consolidation of soil beneath shallow foundation; consolidation of soil beneath deep foundation; settlement of unsaturated cohesive soil. *(Lectures: 3)*
- *Foundation of expansive soil:* expansion potential; basic expansive soil principles; prediction of foundation movement; foundation design for expansive soil; flatwork; expansive rock. *(Lectures: 3)*
- *Retaining walls:* simple retaining wall without wall friction; simple retaining wall with friction; design and construction of retaining walls; restrained retaining walls; mechanically stabilized earth retaining walls; sheet pile walls; temporary retaining walls; moisture migration through basement walls. *(Lectures: 4)*
- *Foundation deterioration and cracking:* timber decay; sulfate attack on concrete; frost; historic structures; shrinkage cracking; moisture migration through slab-on-grade foundations. *(Lectures: 4)*
- *Geotechnical earthquake engineering:* seismological concepts; ground motion parameters; dynamic soil behavior; ground response analysis; liquefaction; slope stability. *(Lectures: 4)*
- *Seismic strengthening/retrofitting methods for foundations:* earthquake structural damage; foundation settlement; retaining walls; foundation alternatives to mitigate earthquake effects. *(Lectures: 4)*
- *Grading and other soil improvement methods:* grading; compaction; soil improvement methods. *(Lectures: 4)*
- *Foundation excavation, underpinning, and field load tests:* foundation excavation and construction; field load tests; foundation underpinning; observational method. *(Lectures: 4)*

- *Subsurface investigation and instrumentations:* traditional in-situ investigations; advanced geophysical tests; field instrumentations. *(Lectures: 3)*
- *International building codes for foundations:* general regulations for footing and foundations; foundation adjacent to slopes; retaining walls. *(Lectures: 3)*

Text Books

1. Bowles JE (1997) Foundation Analysis and Design (5th Ed). Tata-McGraw Hill International Edition. ISBN 0-07-912247-7.
2. Robert W. day (2010). Foundation Engineering Handbook (2nd Ed). McGraw-Hill Professional. ISBN 978-0-07-174010-4.
3. Budhu. M (2012). Soil Mechanics and Foundations (3 rd Ed). John Wiley and Sons. ISBN: 978-0-470-55684-9.
4. Robert W. day (2010). Foundation Engineering Handbook (2 nd Ed). McGraw-Hill Professional. ISBN 978-0-07-174010-4.Manjriker

Additional references:

Foundation Engineering (2018) by Prof. Kousik Deb on NPTEL.
https://nptel.ac.in/courses/105105176/

Title	**Reliability Engineering**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite	Nil		

Objectives: This course aims at:

1. Introducing basic principles of statistics, uncertainty and reliability analysis.
2. Enhancing engineering knowledge and specialist techniques to prevent/reduce the likelihood of failures of structures.
3. Introduction of different methods of reliability analysis in Civil Engineering.

Learning Outcomes: Students will have the ability to:

1. Analyze the statistical moments of input random variables and the output performance function.
2. Determine the probability of failure, reliability index and safety factors of limit state functions.
3. Analytical knowledge of different levels of reliability used for structures.
4. Perform reliability based design on different limit states.
5. Apply finite element based reliability analysis to structural components and systems.

Contents:

- *Introduction to Reliability Engineering:* Concepts of structural safety, limit state and reliability; types of uncertainty; reliability analysis and its importance. *(Lectures: 2)*
- *Basics of statistics and probability*: Basic statistics; theory of probability; probability distribution function; hypothesis testing. *(Lectures: 3)*
- *Resistance parameters and distributions:* Statistics of material properties and strength; dimensional variations; allowable stresses. *(Lectures: 3)*
- *Simulation based study of structural safety:* Monte Carlo method; Latin hypercube sampling; advanced sampling techniques. *(Lectures: 3)*
- *Level 2 reliability methods:* Basic variables and failure surface; first order second moment method. *(Lectures: 3)*
- *Gradient based methods:* first and second order methods; independent, correlated, normal and non-normal random variables based limit states. *(Lectures: 5)*
- *Response surface based methods:* Conventional response surface method; advanced response surface methods with examples. *(Lectures: 4)*
- *Reliability based design:* Stochastic models of loads; determination of partial safety factors; code calibration; safety checking format. *(Lectures: 4)*
- *Reliability of structural systems:* System reliability – series and parallel; bounds on system reliability; modelling of structural systems. *(Lectures: 3)*
- *Advanced reliability methods:* Importance sampling method; subset simulation method; stochastic response surface method. *(Lectures: 3)*
- *Brief on advanced topics:* Fatigue reliability estimation; S-N curve approach; time-dependent reliability. *(Lectures: 3)*
- *Design optimization:* Introduction to the concept of design optimization; examples on solving reliability based design optimization problems. *(Lectures: 3)*
- *Application of reliability analysis and design:* applications to steel, concrete, offshore structures etc. to determine probability of failure and safe design. *(Lectures: 3)*

Text Books:

1. P. Thoft-Christensen and M.J. Baker (1982) Structural Reliability: Theory and its Applications, Springer-Verlag Berlin, Heidelberg. ISBN 978-3-642-68699-3.

2. A.H.S. Ang and W. H. Tang (1984) Probability Concepts in Engineering Planning and Design Vol. II Decision, Risk and Reliability, John Wiley and Sons, New York. ISBN 978-0-471-03201-4
3. P. Thoft-Christensen and Y. Murotsu (1986) Applications of Structural Systems Reliability Theory, Springer Verlag Berlin, Heidelberg. ISBN 978-3-642-82766-2.
4. O. Ditlevsen and H.O. Madsen (1996) Structural Reliability Methods, John Wiley & Sons Ltd, Chichester. ISBN 978-0-471-96086-7.
5. S.-K. Choi, R.V. Grandhi and R.A. Canfield (2007) Reliability-based Structural Design, Springer-Verlag, London. ISBN 978-1-84628-444-1.
6. R. Ranganathan (2011) Structural Reliability Analysis and Design (3rd Ed), Jaico Publishing House, Mumbai. ISBN 81-7224-851-2.
7. R.E. Melchers and A.T. Beck (2018) Structural Reliability Analysis and Prediction (3rd Ed), John Wiley and Sons Ltd, Hoboken, NJ. ISBN 978-1-119-26599-3.

Additional References:

1. Reliability Engineering (2009) by Prof. G.L. Sivakumar Babu on NPTEL. https://nptel.ac.in/courses/105/108/105108128/
2. Structural Reliability (2014) by Dr. Arunasis Chakraborty on NPTEL. https://nptel.ac.in/courses/105/103/105103140/
3. Risk and Reliability of Offshore Structures (2016) by Prof. Srinivasan Chandrasekaran on NPTEL. https://nptel.ac.in/courses/114/106/114106041/

Title	**Structural Dynamics**	Number	CIE7XX0
Department	Civil and Infrastructure Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	B. Tech, M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite	Nil		

Objectives: This course aims at:

1. Introducing the behaviour of structures (especially framed building) to various dynamic loads, *viz.* earthquake, machine vibration etc.
2. Fundamental understanding to analyze structures subjected to dynamic excitation and computing quantities like displacements, forces, stresses etc.
3. Analytical and numerical methods as well as procedures in a way that emphasize physical insight.
4. Problem solving ability for dynamic response in Civil Engineering design.

Learning Outcomes: Students will have the ability to:

1. Establish dynamic equilibrium and equation of motion for structural systems.
2. Analyze the dynamic response and stress generation in single and multi degree of freedom systems for different types of loading.
3. Implement the basis for the classical energy formulations in dynamical systems.
4. Model structural damping and solve problems in frequency as well as time domain.
5. Solve the eigenvalue problems and gain knowledge on its useful properties.
6. Device methods to protect new and existing structures against seismic collapse.
7. Apply the structural dynamics theory to real-world problems like seismic analysis and design of structures.

Contents:

- *Basics of structural dynamics:* Introduction to dynamic loading and response; types of analysis - static and dynamic loads; degrees of freedom; lumped mass; types of mass-spring-damper system; formulation of equation of motion for structural dynamics using direct equilibration, principle of virtual displacement and Hamilton's principle. *(Lectures: 4)*
- *Free vibration of single degree of freedom (SDOF) system:* Undamped free vibration and its solution; natural period and frequency; energy in free vibration; damped free vibration; types of damping; concept of logarithmic decrement. *(Lectures: 4)*
- *Forced vibration of SDOF system:* Undamped forced vibration and its solution; amplitude and phase angle; dynamic amplification factor; damped forced vibration. *(Lectures: 3)*
- *Force transmission and vibration measurement:* Resonant frequency and half power band width; force transmission and isolation; design of vibration measuring instruments. *(Lectures: 3)*
- *Response to harmonic and periodic motions:* Response of SDOF systems to harmonic loads and periodic excitations; Fourier series representation. *(Lectures: 4)*
- *Response to arbitrary motions:* Response of SDOF systems to unit impulse excitation, arbitrary force (solution using Duhamel's integral), step and ramp forces, rectangular pulse and half sinusoidal waves. *(Lectures: 4)*
- *Numerical methods of solution:* Determining the dynamics response using methods such as time stepping methods, central difference method and Newmark's method. *(Lectures: 4)*
- *Response spectrum:* Concept of response spectrum; uses of response spectrum; complex frequency transfer function; development of tripartite plot; base shear and bending moment; response of structure in frequency domain. *(Lectures: 4)*
- *Multi-degree of freedom (MDOF) systems:* Equation of motion for MDOF system; solution of equation, matrix properties; natural frequencies and mode shapes; static condensation; modal orthogonality; mode superposition analysis of MDOF system; approximate method for finding natural frequency. *(Lectures: 5)*

- *Earthquake response of MDOF systems:* Time history analysis; response spectrum analysis. *(Lectures: 2)*
- *Dynamic response of continuous systems:* Vibration of continuous systems; axial vibration of prismatic bar; flexural vibration of beams; shear and bending behaviour; generalized SDOF. *(Lectures: 3)*
- *Applications:* Introduction to finite element method; formation of element stiffness, mass and force matrices; examples for seismic analysis and response. *(Lectures: 2)*

Text Books:

1. J.M. Biggs (1964) Introduction to Structural Dynamics, McGraw Hill, NY. ISBN 07-005255-7.
2. R.W. Clough and J. Penzien (1993) Dynamics of Structures (2nd Ed), McGraw Hill Inc Editor. ISBN 0-07-011392-0.
3. A.K. Chopra (2017) Dynamics of Structures Theory and Application to Earthquake Engineering (5th Ed), Prentice Hall, NJ. ISBN 978-0-134-55512-6.
4. A.K. Jain (2017) Dynamics of Structures with MATLAB® Applications, Pearson Education, Chennai, India. ISBN 978-9-332-55855-7
5. M. Paz and Y.H. Kim (2019) Structural Dynamics Theory and Computation (6th Ed), Springer Nature, Switzerland. ISBN 978-3-319-94742-6.

Additional References:

1. Structural Dynamics (2012) by Dr. Pradipta Banerji on NPTEL. https://nptel.ac.in/courses/105/101/105101006/
2. Structural Dynamics (2016) by Prof. Ramancharala Pradeep Kumar on NPTEL. https://nptel.ac.in/courses/105/106/105106151/
3. Structural Dynamics for Civil Engineers - SDOF Systems (2019) by Prof. Riya Catherine George on NPTEL. https://nptel.ac.in/courses/105/104/105104189/

Title	**Advanced Mechanics of Solids**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P-D [C]	3-0-0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

1. Enable students to learn advanced concepts of stress analysis
2. Apply concepts to analysis of mechanical systems such as rotating disks, thick cylinders etc.

Learning Outcomes

1. Develop Ability to solve advanced stress analysis problems using analytical and numerical methods
2. Ability to formulate and solve 3D problems in stress analysis

Contents

1. Introduction to Cartesian tensors, Stress: Normal and shear stress, Equilibrium equations (force and moment), Symmetry of shear tensor, Equilibrium at boundary (Cauchy Equation). 2-D problems Plane stress, Stress transformation, Principal stresses, maximum shear stress, Mohr's circle *(7 Lectures)*
2. Displacements and strains, Green's strain, Compatibility condition, Plane strain Problems, Strain transformation, Strain gage and rosette *(5 Lectures)*
3. Structural Approximations: Physical and mathematical. Structural elements-bars, beams, plates, membrane, shear panel, shells *(1 Lectures)*
4. Mechanical behavior of Engineering Materials, Effect of Temperature, Anisotropic, Isotropic, Orthotropic materials, Failure Criteria (Distortion strain energy, Octahedral shear stress) Maximum normal stress, Maximum Shear stress (Tresca), Maximum distortion energy (von-Mises and Hencky) *(4 Lectures)*
5. Theory of Elasticity, Plane Stress Problem with Stress formulation in 2-D, Plane Strain Problem with Stress Formulation in 2-D, Airy's stress function, stress, strain and equilibrium in polar coordinates, Displacement formulation with plane stress, St. Venant's Principle *(4 Lectures)*
6. Bending and extension of beams, stress-resultants, Equilibrium equations in terms of deformation with transverse and axial load *(3 Lectures)*
7. Torsion of slender bodies (beams), Prandtl stress function formulation, Membrane analogy, Closed form beam torsion analytical solution, Warping function formulation, Thin Walled open sections, Torsion constants for rolled and extruded beams, Thin walled closed sections, Shear flow *(8 Lectures)*
8. Beam shearing stresses due to shear forces, Shear Centre *(3 Lectures)*
9. Euler Buckling *(3 Lectures)*
10. Principle of virtual-work *(4 Lectures)*

Text Books

1. Srinath, L.S. (2009), Advanced Solid Mechanics, Tata McGraw-Hill
2. Timoshenko, S.P., and Goodier, J.N. (1980), Theory of Elasticity, Tata McGraw- Hill
3. Sadd, M. H. (2006), Elasticity: Theory, Applications and Numerics, Academic Press

Online course Material

1. http://nptel.ac.in/courses/105106049/

Title	**Substation Infrastructure**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P [C]	2–0–0 [2]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives

The Instructor will:

1. Provide exposure to layout of various components of substation and design its infrastructure.

Learning Outcomes

The students are expected to have the ability to:

1. Develop the layout of a substation and design the substation infrastructure

Contents

Substation Layout: Substation design considerations, layouts - various bus configurations and space requirements (5 lectures)
Auxiliary Power: Battery and charger configurations, battery capacity, battery sizing calculations, AC power supplies, auxiliary transformation connections, earthing transformer (5 lectures)
Current and Voltage Transformers: Types, design and construction constructions, (4 lectures)
Insulators: Insulator materials, types, design consideration, testing (4 lectures)
Earthing and Bonding: Touch and step voltage, earthing materials, earth resistance, earth potential rise, protective multiple earthing (5 lectures)
Switchgear: Introduction, switches, disconnectors, contactors, introduction to operating mechanism and arc quenching (5 lectures)

Textbook

1. Bayliss,C., Hardy,B., (2012), *Transmission and Distribution Electrical Engineering*, 4th Edition, Newnes publications
2. Rao,S., (1992), *Electrical Substation Engineering and Practice Engineering & Practice EHV-AC, HVDC AND SF_6-GIS*, 3rd Edition, Khanna publications

Title	**Gas Insulated Substation**	Number	**EE7XX0**
Department	Electrical Engineering	L-T-P [C]	2–0–0 [2]
Offered for	M. Tech, M. Tech-Ph.D, Ph.D	Type	ME
Prerequisite			

Objectives

The Instructor will:

1. Provide exposure to of gas insulated substations (GIS), its design, testing, and diagnostics methods

Learning Outcomes

The students are expected to have the ability to:

1. Design and test a gas insulated substation

Contents

Introduction: Features, economics, arrangement, and components of gas insulated substations, comparison with air insulated substations (4 lectures)

Introduction to SF_6: physical, chemical, and electrical properties, handling, specifications for GIS applications (3 lectures)

Design of GIS: Ratings of components, estimation of electric stress, design aspects of GIS components, insulation design, insulation coordination, grounding systems (6 lectures)

Testing: GIS testing, on-site tests, condition monitoring and diagnostic methods (4 lectures)

Problems and Diagnostics: Particles and their effects, SF_6 gas decomposition, characteristics of imperfections in insulation, insulation diagnostics methods, partial discharge in GIS and its detection (8 lectures)

Fast transient phenomena: Origin, characteristics, and effects of very fast transient overvoltages (3 lectures)

Textbook

1. Naidu,M.S., (2008), *Gas Insulated Substations*, I.K. International Publishing House Pvt Ltd

2. Koch,H., (2014), GIS *Gas Insulated Substations*, IEEE Press and John Wiley & Sons Ltd

Title	**Machine Learning-1**	Number	**CS 7XX0**
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite	None		

Objectives: The Instructor will:

1. Provide motivation and understanding of the need and importance of Machine Learning in today's world.
2. Provide details about various algorithms in Machine Learning

Learning Outcomes: The students are expected to have the ability to:

1. Develop a sense of Machine Learning in the modern context, and independently work on problems relating to Machine Learning.
2. Design and program efficient algorithms related to Machine Learning, train models, conduct experiments, and deliver ML-based applications

Contents

Fractal 1

- Introduction: Motivation, Different types of learning, Linear regression, Logistic regression (2 lectures)
- Gradient Descent: Introduction, Stochastic Gradient Descent, Subgradients, Stochastic
- Gradient Descent for risk minimization (2 lectures)
- Support Vector Machines: Hard SVM, Soft SVM, Optimality conditions, Duality, Kernel trick, Implementing Soft SVM with Kernels (4 lectures)
- Decision Trees: Decision Tree algorithms, Random forests (2 lectures)
- Nearest Neighbour: k-nearest neighbour, Curse of dimensionality (1 lecture)
- Neural Networks: Feedforward neural networks, Expressive power of neural networks, SGD and Backpropagation (3 lectures)

Fractal 2

- Clustering: Linkage-based clustering algorithms, k-means algorithm, Spectral clustering (2 lectures)
- Dimensionality reduction: Principal Component Analysis, Random projections, Compressed sensing (2 lectures)
- Generative Models: Maximum likelihood estimator, Naive Bayes, Linear Discriminant Analysis,
- Latent variables and Expectation-maximization algorithm, Bayesian learning (4 lectures)
- Feature Selection and Generation: Feature selection, Feature transformations, Feature learning (3 lectures)
- Model selection and validation: Validation for model selection, k-fold cross-validation,
- Training-Validation-Testing split, Regularized loss minimization (3 lectures)

Fractal 3

- Statistical Learning Framework: PAC learning, Agnostic PAC learning,
- Bias-complexity tradeoff, No free lunch theorem, VC dimension, Structural risk minimization, Adaboost (7 lectures)
- Foundations of Deep Learning: DNN, CNN, RNN, Autoencoders (7 lectures)

Textbook

1. Shalev-Shwartz,S., Ben-David,S., (2014), Understanding Machine Learning: From Theory to Algorithms, Cambridge University Press.

Reference books:

1. Mitchell Tom (1997). Machine Learning, Tata McGraw-Hill.

Self-Learning Materials

1. Department of Computer Science, Stanford University, https://see.stanford.edu/Course/CS229

Title	**Artificial Intelligence – 1**	Number	**CS 7XX0**
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M. Tech, M. Tech-PhD, Ph.D	Type	ME
Prerequisite			

Objectives: The Instructor will:

1. Cover various paradigms that come under the broad umbrella of AI, with some of them being covered in depth

Learning Outcomes: The students are expected to have the ability to:

1. Develop an understanding of where and how AI can be used

Contents

- Introduction (1 lecture)
- Propositional logic (8 lectures)
- Search: Uninformed strategies (BFS, DFS, Dijkstra), Informed strategies (A* search, heuristic functions, hill-climbing), Adversarial search (Minimax algorithm, Alpha-beta pruning) (10 lectures)
- Predicate logic: Knowledge representation, Resolution (6 lectures)
- Rule-based systems: Natural language parsing, Context free grammar (3 lectures)
- Constraint satisfaction problems (4 lectures)
- Planning: State space search, Planning Graphs, Partial order planning (4 lectures)
- Uncertain Reasoning: Probabilistic reasoning, Bayesian Networks, Dempster-Shafer theory,
- Fuzzy logic (6 lectures)

Textbook

1. Russel,S., and Norvig,P., (2015), Artificial Intelligence: A Modern Approach, 3rd Edition, Prentice Hall.

Reference books:

1. Research literature

Self-Learning Materials

1. Department of Computer Science, University of California, Berkeley, http://www.youtube.com/playlist?list=PLD52D2B739E4D1C5F
2. NPTEL: Artificial Intelligence, https://nptel.ac.in/courses/106105077/

Title	**Wave Energy Conversion**	Number	
Department	Chemical Engineering	L-T-P-D [C]	1-0-0 [1]
Offered for	M.Tech, M.Tech.-Ph.D. [CIE]	Type	ME
Prerequisite			

Objectives: this course aims at:

1. Introducing elements of wave energy, wave energy converters (WECs), conversion systems, its economics, and future potential.

Learning Outcomes: Students will have the ability to

1. Understand the geographical distribution of wave energy potential.
2. Classify wave energy conversion process, and choose different near-shore to offshore conversion devices.
3. Understand principles of different components of power conversion systems.
4. Develop cost & performance analysis of shoreline, near-shore, and offshore systems.
5. Understand positive and negative impacts of such systems and methods to minimize them.

Contents

- ***Ocean waves Wave and Wave Energy:*** Origin and description of waves; Wave climatology and resource distribution; Wave energy resource assessment and development; Wave power activities across the globe.
- ***Wave energy conversion system and power transfer systems:*** Conversion technology; conversion development; Mechanical interface; energy storage; Electric power conversion; Transmission to grid
- ***Economics:*** Cost of wave energy; capital, operating, and generating cost; economics of wave-powered desalination.
- ***Environmental and socioeconomics implications:*** Benefits; Impacts; Impact assessment; Other considerations.

Text Book

1. Bhattacharyya R, McCormick ME (2003). Wave Energy Conversion. Volume 6. Elsevier Ocean Engineering Book Series. Elsevier. ISBN: 0 08 044212 9

Reference Books

1. Arthur Pecher, Jens Peter Kofoed, (2017) Handbook of Ocean wave Energy, Springer
2. T.W. Thorpe, (1999) A Brief Review of Wave Energy, UK Department of Trade and Industry, ETSU-R120, 25 May 99

Online Material:

1. https://nptel.ac.in/content/storage2/courses/108108078/pdf/chap8/teach_slides08.pdf